Optimal Unconventional Policy in a New Keynesian Preferred Habitat Model*

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Abstract

We study the transmission mechanisms of monetary policy in a general equilibrium model featuring heterogeneous households, nominal rigidities, and limits to arbitrage. The interaction of bond market segmentation and household heterogeneity introduces novel rebalancing transmission mechanisms of monetary policy, rationalizing a number of empirical puzzles regarding the reactions of term premia to short-term interest rates. Term premia volatility enters the social welfare function through an imperfect risk-sharing channel; conservative calibrations imply costs on par with traditional stabilization motives. Interest rate smoothing can improve welfare, but necessarily comes at the cost of increased macroeconomic volatility. With balance sheet tools, we derive a separation principle for optimal policy: conventional policy stabilizes aggregate fluctuations, while unconventional policy stabilizes term premia. Balance sheets are necessary but imperfect tools for macroeconomic stabilization when short rates are constrained. Our model provides a justification and operational guidance for a central bank "triple mandate" which seeks to stabilize output, inflation, and long-term rates.

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1 Introduction

Central banks responded aggressively to worsening financial conditions and growing recessionary pressure during the global financial crisis of 2007-8. In addition to steep cuts in policy rates, central banks undertook various unconventional policy actions; the most salient of these were the quantitative easing (QE) programs carried out by the Federal Reserve. With a binding lower bound on the federal funds rate, the purpose of these policies was to reduce long-term rates for households and firms in order to stimulate the economy. The Fed continued to utilize QE programs during the onset of COVID-19, and then began implementing quantitative tightening (QT) in response to growing inflationary pressure starting in 2022.

The purpose of this paper is to study the positive and normative consequences of unconventional monetary policy, and how the design of balance sheet policies interacts with the conduct of more conventional interest rate policies. We develop a tractable general equilibrium model with market segmentation, financial frictions, nominal frictions, and household heterogeneity. We start with a conventional New Keynesian model, where producers face nominal rigidities when setting prices. However, we relax the assumption of frictionless financial markets: households and firms face imperfect bond market access.

We explicitly model bond markets and the determination of the entire term structure; in particular, we embed within our dynamic general equilibrium model a segmented bond market which builds on the seminal paper of Vayanos and Vila (2021). Households and firms have differentiated access to financial assets of different maturities. The positions these agents take in bond markets induce "preferred habitat" market segmentation forces across the term structure. Specialized bond intermediaries re-integrate markets; but when arbitrageur risk-bearing capacity is imperfect, this integration is only partial, and the endogenous price of risk matters for both aggregate dynamics and welfare.

Relative to representative agent models, our setup introduces the possibility of imperfect risk-sharing across households. Household consumption and savings decisions now take place across the entire term structure of bond returns. If bond arbitrageurs have perfect risk-bearing capacity, this friction is immaterial in equilibrium, but outside of this special case, borrowing rates differ across differentiated households, inducing inefficient consumption dispersion across households.

We show that these bond market frictions have important implications for how monetary policy transmits to households and the aggregate economy. First, consider the key mechanisms of conventional monetary policy. Changes in the policy (short-term) interest rate are transmitted to households only via segmented bond markets. In particular, household and firm demand for long-term bonds implies portfolio rebalancing on the part of bond arbitrageurs; the risk exposure of arbitrageurs therefore changes in response to short rate movements. Imperfect risk-bearing capacity therefore implies fluctuations in term premia, and so a change in the policy rate is not transmitted one-for-one to all borrowing rates, but instead differs across the term structure and therefore across the portfolios of differentiated households.

Next, consider unconventional (QE or QT) policies. Central bank asset purchases and sales directly induce portfolio rebalancing among bond market investors. Once again, when risk-bearing capacity of bond markets is imperfect, such rebalancing implies changes in term premia, even when the policy rate is unchanged. Because households borrow across the term structure, these policies also affect household consumption decisions.

Our general equilibrium model uncovers novel transmission mechanisms of monetary policy. Through market clearing, the bond market positions across heterogeneous households are crucial for understanding movements in term premia. But movements in term premia themselves induce relative changes in bond market positions across the entire distribution of households. Nevertheless, we characterize the dynamics of the model as a function of a small number of sufficient statistics of the household asset distribution. Intuitively, term premia are a function of the equilibrium exposure of bond arbitrageurs to the sources of aggregate risk in the economy. Market clearing implies that the entire distribution of bond positions across firms and households is necessary to pin down arbitrageur bond holdings; however, it is only the *risk-weighted distribution* across the term structure which is relevant for the pricing of risk in equilibrium. Because there are only a small number of aggregate risk factors, we show that the dynamics of the risk-weighted distribution itself is summarized by a small number of factors.

The model makes clear predictions for the reaction of term premia to movements in short-term interest rates over different time horizons, and as a function of the conduct of monetary policy. We show that when firm borrowing demand falls substantially during recessionary shocks (which occurs under reasonable parameterizations), the

rebalancing mechanism implies large reductions in term premia on impact. However, this induces additional borrowing motives for households exposed to long-duration assets. In particular, slow-moving portfolio rebalancing of households implies that intermediaries become more exposed to long-duration assets. Thus, over longer horizons, the reaction of term premia becomes small or even positive. We further show that when the short rate is constrained and monetary policy is conducted using balance sheet tools, the reaction of term premia is larger. However, in response to cuts in short rates due to monetary policy *shocks* which imply modest increases in firm borrowing demand, the reaction of term premia is smaller, or may even reverse sign. Following the empirical methodology of Hanson et al. (2021), we find strong support for our model predictions in US Treasury markets: (i) unconditionally, increases in short-term rates are associated with large increases in long-dated forward rates over short-horizons; (ii) this reaction is substantially larger during the QE period; (iii) but over longer horizons, increases in short-term rates are associated with small increases or even decreases in long-dated forward rates; (iv) the conditional reaction of long-dated forward rates to high-frequency-identified monetary policy shocks are smaller.

Next we turn to the welfare implications of our model. From the perspective of an "aggregate Euler equation" channel of monetary policy, our model thus implies that conventional and unconventional policies are somewhat substitutable: either policy can be used to target borrowing rates faced by households. In fact, conditional on the dynamics of an appropriately defined "effective rate" composed of the weighted average of bond returns across the term structure, the first-order aggregate dynamics of output and inflation in our model are the same as a model in which a representative household borrows at this effective rate. This implies that if the central bank loss function only depends on the volatility of output and inflation, both conventional and unconventional policies can achieve identical outcomes. In particular, if "divine coincidence" holds, then either policy tool can achieve first-best.

However, we show that such a policy loss function is *suboptimal* from a social welfare perspective. Both policies lead to variation in term premia, and excess fluctuations in term premia implies excess dispersion in borrowing rates. Fluctuations across the term structure imply differentiated consumption and savings decisions across households. This dispersion results in utility losses relative to the first-best because of imperfect risk-sharing. Even when "divine coincidence" holds, we show

that neither tool alone can achieve first-best. Quantitatively, we show that even under conservative assumptions, the heterogeneity in risk exposure across US households along with the volatility of term premia imply that this welfare motive is of the same order of magnitude as (and plausibly larger than) standard aggregate stabilization motives.

We therefore derive the optimal mix of policy rules when the central bank is maximizing social welfare. When short-rate and balance policy tools are unconstrained, we derive an optimal separation result: conventional policy targets the stability of inflation and the output gap, while unconventional policy targets stability of term premia. When divine coincidence holds, this policy achieves first-best.

However, when policy constraints bind, policy must balance tradeoffs. First, if the central bank faces balance sheet constraints, optimal policy implies that the short rate must be less reactive to aggregate shocks. This is because of the endogenous term premia volatility channel: highly reactive conventional policy endogenously generates more volatile bond returns which worsens risk-sharing. However, this necessarily comes at the cost of increased macroeconomic instability. Second, if the central bank faces constraints on short-rate policy, then QE must be used to offset macroeconomic shocks. However, this policy has welfare consequences: term premia are more volatile than in the first-best, and thus consumption dispersion causes social welfare losses. Third, when divine coincidence does not hold, when firm and household borrowing rates differ, it is possible for the policymaker to utilize short rate and balance sheet policies simultaneously to achieve full macroeconomic stabilization. However, the welfare costs of such a policy are large as such policies necessarily involve large term premia manipulations; in fact, we show the welfare costs become unboundedly large when firm and household borrowing rates converge.

Thus, policy constraints imply tradeoffs between macroeconomic stabilization motives and risk-sharing motives. Our findings apply when the central bank pursues simple time-consistent policy rules (where policy tools are functions of the natural state variables only). We also derive optimal policy results when the central bank has full commitment and can choose policy tools freely as a function of current and past realizations of the economy. Such policies are welfare-improving over simple policy rules. For instance, when the central bank can only utilize conventional policy, optimal policy under full commitment implies interest rate changes are smoothed out relative to the optimal time-consistent short rate rule. The ability to commit to

further smoothing of interest rate changes implies a reduction in short rate volatility; in equilibrium, term premia are smaller and less volatile. Relative to simple time-consistent policy rules, optimal commitment induces inertia for an additional novel reason: inertial policy rules imply *ex-ante* smaller expected variation in policy tools, which in equilibrium implies smaller required risk compensation and therefore improves risk-sharing.

Additionally, while it is not our main focus, our paper makes broader methodological contributions: (i) our perturbation approach (with first-order endogenous risk premia) allows consumption dispersion and risk-sharing motives to enter social welfare with the same order as aggregate volatility welfare at the same order as output and inflation; (ii) the continuum of households with heterogeneous wealth is collapsed through its interaction with risk pricing into a small number of sufficient statistics; (iii) the characterization of the full-commitment policy remains finite-dimensional even though policy explicitly manipulates endogenous volatilities across the continuum of bonds. Our paper therefore has wider applicability for heterogeneous agent models with financial frictions.

A general message of our model is that implementation matters for welfare. Consumption dispersion depends directly on the entire term structure of bond premia, which in turn are intimately related to the expected path and volatility of the short-term interest rate. Thus, it is not possible to characterize optimal policy purely as a function of the path of aggregate variables like the output gap and inflation, as these variables alone are not sufficient to characterize the term structure of interest rates.

Finally, our model implies that the goal of optimal policy is characterized by a "triple mandate" of stable inflation, output, and long-term bond returns. This seems to conflict with the commonly understood "dual mandate" of the Federal Reserve. However, the Congressional mandate of the Federal Reserve is to "promote effectively the goals of maximum employment, stable prices, and moderate long-term interest rates." Standard interpretations of both academics, practitioners, and policymakers of this legal mandate is in terms only of the first two objectives; the theoretical reasoning is that stabilizing the macroeconomy necessarily implies moderate long-term yields.[1] Our model shows that macroeconomic stabilization is of course a key policy objective of the central bank, but is not always sufficient for stabilizing long-term rates. Thus, our paper provides a theoretical justification for why the triple mandate of the Federal

[1]For further discussion, see Mishkin (2007), "Monetary Policy and the Dual Mandate."

Reserve is optimal,[2] and provides guidance for the implementation of a policy with three distinct mandates.

Our paper builds on the seminal preferred habitat model of Vayanos and Vila (2021), which formalizes the original concept as described in Modigliani and Sutch (1966).[3] The main insight of preferred habitat models is that the interaction of clientele investors implies important departures from the expectations hypothesis in the determination of the term structure of interest rates. More concretely, demand and supply shocks in these markets induce changing risk exposure on the part of marginal bond investors; when risk-bearing capacity is imperfect, this implies fluctuations in risk premia.[4] These models are typically partial equilibrium models. One exception is Ray et al. (2024), which uses a quantitative version of the model in this paper to study the positive implications of QE policies. That paper considers a richer risk factor structure and a wider set of assets (both riskless and risky bonds), but households are representative; hence, the dynamics in this paper induced by household heterogeneity are not present, and the normative implications explored are absent.

Our work more generally contributes to a large literature studying the effects of QE in general equilibrium models. These models feature various forms of financial frictions such as bank balance sheet constraints which break the textbook "QE neutrality" results (e.g. see Gertler and Karadi 2011, Gertler and Karadi 2013, Cúrdia and Woodford 2011, Chen et al. 2012, Sims and Wu 2020, Karadi and Nakov 2020, Iovino and Sergeyev 2023, Carlstrom et al. 2017, Ippolito et al. 2018, Alexandrov and Brunnermeier 2026).

Our paper also relates to models of market segmentation which study various forms of macroeconomic stabilization or macro-prudential policy (e.g. see Andrés et al. 2004, Cui and Sterk 2021, Auclert 2019, Angeletos et al. 2023, Debortoli and Galí

[2]This is true under the interpretation that "moderate long-term rates" implies stabilizing excessive fluctuations in term premia. An alternative view of the third mandate is that the central bank should seek to keep long-term borrowing rates "low" for reasons related to the sustainability of fiscal debt. Such an interpretation is not valid in our model, and indeed would be suboptimal.

[3]Other important theoretical contributions to preferred habitat models are Greenwood and Vayanos (2014), Greenwood et al. (2016), King (2019b), King (2019a), Kekre et al. (2024).

[4]Empirically, there is strong evidence of the existence of demand and supply "preferred habitat" frictions considered in this paper, and that these frictions are important for understanding the transmission of large-scale asset purchases (e.g. see Krishnamurthy and Vissing-Jorgensen 2011, Hamilton and Wu (2012), D'Amico and King 2013, Li and Wei 2013, Krishnamurthy and Vissing-Jorgensen 2012, Cahill et al. 2013, King 2019b, Fieldhouse et al. 2018, Di Maggio et al. 2020, Debortoli et al. 2020).

2017). Our findings on both the transmission and the optimal design of conventional policy are closely related to the results in papers such as Caballero and Simsek (2024) and Caballero et al. (2024): the transmission of monetary policy to the relevant borrowing rates for macroeconomic dynamics is imperfect; and more fundamentally, depends on the systematic policy rule as this is what is relevant for the endogenous pricing of risk.

Finally, our model overlaps with similar work in an international setting. Most closely related to our optimal policy results is Itskhoki and Mukhin (2023), who study the optimal mix of conventional policy and FX interventions in an open economy setting. Theoretically more closely linked to our framework, Gourinchas et al. (2022) and Greenwood et al. (2023) also study the determination of bond yields and exchange rates in a similar preferred habitat model, though these papers do not study general equilibrium effects.

Our paper is structured as follows. Section 2 describes the private agents in the model, characterizes general equilibrium, and derives the social loss function which the policymaker seeks to minimize. Section 3 studies the aggregate dynamics analytically in a simple version of the model with ad-hoc policy rules. Section 4 empirically tests the key predictions of the model regarding the dynamics of short-term rates and term premia. Section 5 characterizes optimal policy as a function of different assumptions regarding constraints on balance sheet tools and short-term rates, while Section 6 extends these results to the case of full commitment. Section 7 discusses additional extensions and tests of our model, and Section 8 concludes.

2 Model

Time is continuous and denoted by $t \in [0, \infty)$. The model features relatively standard firm and household sectors. The household sector is formally comprised of differentiated households making labor, consumption, and savings decisions. Intermediate goods are produced by monopolistically competitive firms using labor; these firms set prices but face nominal rigidities in the form of Rotemberg pricing frictions. Differentiated goods are aggregated by a perfectly competitive final goods retail sector.

There is a continuum of zero-coupon bonds with maturity $\tau \in [0, T]$: a τ-maturity bond has price $P_t^{(\tau)}$ at time t and pays one dollar at maturity date $t + \tau$. Both firms and households face financial frictions when trading bonds. Each household makes

borrowing and savings decisions, but must trade with a bond fund which has fixed portfolio weights across bonds. This friction captures a number of realistic features of household portfolio decisions. The different portfolio weights across households represent a type of preferred habitat for households (for instance, due to demographics such as age) which we summarize parsimoniously as heterogeneity in duration exposure. In reality, this reflects for example some households who own property which is financed with long-duration mortgage securities; while for others, saving decisions are predominantly in short-duration securities (such as money market mutual funds or simple bank deposits). Further, our assumption also reflects the fact that households typically do not directly buy and sell financial assets but instead interact with pension funds and mutual funds, which tend to allocate their portfolio using fixed weights and rebalance sluggishly (e.g., Koijen and Yogo (2019), Koijen and Yogo (2022), Bretscher et al. (2022), Peng and Wang (2022)). In the spirit of the "preferred habitat" literature, we capture these frictions with the parsimonious but stark assumption that each household faces heterogeneous but fixed portfolio weights. Appendix B.5 discusses microfoundations and extensions. Firms also face a financial friction in the form of a "working capital" constraint: each firm must borrow a portion of its wage bill in advance of production. Analogous to the household portfolio problem, we also assume that the financing costs for firms depend on a portfolio of bonds with fixed portfolio weights.

Because neither households nor firms can freely trade bonds of different maturities, specialized bond traders intermediate bond markets. Bond arbitrageurs trade bonds across the entire term structure, but have limited risk-bearing capacity. These agents are owned by the household sector, but due to financial frictions do not price bonds using the household stochastic discount factor (SDF). We represent the arbitrageur portfolio problem as a function of risk aversion (formally, preferences of the arbitrageurs); however, we treat this risk aversion parameter as a proxy for risk-bearing capacity of financial intermediaries. This is a friction which hinders arbitrageurs' ability to trade assets perfectly on behalf of households as a whole.

The monetary authority sets the short-term nominal interest rate i_t, and conducts balance sheet policies. A fiscal authority provides optimal production subsidies but is otherwise passive and balances the budget period by period via lump-sum taxes.

We study versions of the model where the monetary authority chooses policy according to an ad-hoc Taylor rule, and compare this to policy rules which are chosen

to maximize social welfare (possibly subject to implementation frictions). We focus on a linearized equilibrium and second-order welfare approximations. Our linear-quadratic approximation is around a deterministic first-best steady state, where our approximation method still allows for non-zero bond term premia which affect first-order macroeconomic dynamics.

2.1 Setup

Risk factors. We first specify the notation and assumptions for the set of exogenous risk factors in the model. We assume there exists an N_z-dimensional vector $\mathbf{z}_t$ which captures all exogenous sources of aggregate shocks in the model, and which evolves according to

$$\mathrm{d}\mathbf{z}_t = -\boldsymbol{\kappa}_z \mathbf{z}_t \,\mathrm{d}t + \boldsymbol{\sigma}_z \,\mathrm{d}\mathbf{B}_t \,, \tag{1}$$

where $\mathbf{B}_t$ is an N_z-dimensional vector of independent Brownian motions. In the baseline the only factor is aggregate technology, $Z_t = \bar{Z}e^{z_t}$, $z_t \in \mathbf{z}_t$; more generally $\mathbf{z}_t$ collects household discount-factor, cost-push, firm financing, bond demand, and policy shocks, which we discuss below. Where applicable, we treat $\mathbf{z}_t$ generically.

In equilibrium, bond returns can be written as

$$\frac{\mathrm{d}P_t^{(\tau)}}{P_t^{(\tau)}} = \mu_t(\tau)\,\mathrm{d}t + \boldsymbol{\sigma}_t(\tau)\,\mathrm{d}\mathbf{B}_t \,, \tag{2}$$

with $\mu_t(\tau), \boldsymbol{\sigma}_t(\tau)$ determined in equilibrium. No-arbitrage restrictions imply that the nominal short rate is $i_t \equiv \lim_{\tau\to 0} -\log P_t^{(\tau)}/\tau$ (implemented by the central bank via instantaneously riskless reserves in zero net supply).

Households. The household sector is made up of individuals $i \in [0,1]$. Each period, household i chooses a consumption bundle $C_t(i)$ and supplies labor $N_t(i)$. The per-period flow utility (which is identical across households) is

$$u\left(C_t(i), N_t(i)\right) = \frac{C_t(i)^{1-\varsigma} - 1}{1-\varsigma} - \frac{N_t(i)^{1+\varphi}}{1+\varphi}, \tag{3}$$

where ς is the inverse intertemporal elasticity of substitution and φ is the inverse Frisch labor elasticity.

As discussed above, households face financial frictions and so cannot trade all bonds freely. Instead, they save and borrow through bond funds, which determine the allocation of the portfolio across assets. A household i therefore faces the following budget constraint

$$\mathrm{d}\mathcal{B}_t(i) = [\mathcal{W}_t N_t(i) - \mathcal{P}_t C_t(i)]\,\mathrm{d}t + \mathcal{B}_t(i)\,\mathrm{d}\tilde{R}_t(i) + \mathrm{d}\mathcal{F}_t\,, \tag{4}$$

where $\mathcal{W}_t$ is the nominal wage and $\mathcal{P}_t$ is the price index of consumption bundle (both of which are taken as given and identical across households). Household i has nominal wealth $\mathcal{B}_t(i)$ (with $\mathcal{B}_0(i)$ given), earning the realized instantaneous return

$$\mathrm{d}\tilde{R}_t(i) \equiv \int_0^T \eta_{it}(\tau)\frac{\mathrm{d}P_t^{(\tau)}}{P_t^{(\tau)}}\,\mathrm{d}\tau = \tilde{\mu}_t(i)\,\mathrm{d}t + \tilde{\boldsymbol{\sigma}}_t(i)\,\mathrm{d}\mathbf{B}_t\,. \tag{5}$$

The portfolio weights (normalized $\int_0^T \eta_{it}(\tau)\,\mathrm{d}\tau = 1$) are determined by the bond fund with which household i invests, and are taken as given by the household.[5] Finally, the term $\mathrm{d}\mathcal{F}_t$ represents nominal flow transfers (described below). We denote real wealth by $B_t(i) \equiv \frac{\mathcal{B}_t(i)}{C(i)\mathcal{P}_t}$ (relative to steady state consumption $C(i)$, defined below).

The resulting value function of household i at time $t = 0$ is

$$V_0(i) \equiv \max_{\{C_t(i),N_t(i)\}_{t=0}^{\infty}} \mathbb{E}_0 \int_0^{\infty} e^{-\rho t + \varrho_t} u\left(C_t(i), N_t(i)\right)\mathrm{d}t\,, \tag{6}$$

where ϱ_t is an aggregate discount-factor shock with dynamics given by

$$\mathrm{d}\varrho_t = -\kappa_\varrho \varrho_t\,\mathrm{d}t + \boldsymbol{\sigma}_\varrho\,\mathrm{d}\mathbf{B}_t\,, \tag{7}$$

where using the generic shock vector notation from (1), we have $\varrho_t \in \mathbf{z}_t$. The household problem is to maximize (6) subject to (4).

As shown in Appendix B.1, intratemporal optimality conditions are standard and

[5] The baseline model we maintain the stark assumption that each household can only access bond markets through a specific bond fund. In Appendix B.5, we relax this assumption and instead assume that households may directly trade bonds of different maturities, but face rebalancing costs. Our model is the limit when such "sluggish rebalancing" costs grow, and initial portfolio positions take the form discussed in the baseline model.

given by:

$$W_t = C_t(i)^{\varsigma} N_t(i)^{\varphi} \tag{8}$$

characterizes the consumption and labor decision as a function of the real wage $W_t = \frac{\mathcal{W}_t}{\mathcal{P}_t}$. The intertemporal condition for consumption (in log-deviations from steady state) is also mostly familiar, given by

$$\mathrm{E}_t \, \mathrm{d}c_t(i) = \varsigma^{-1} \left[\tilde{\mu}_t(i) - \pi_t - \kappa_\varrho \varrho_t + \xi_t(i) \right] \mathrm{d}t \,, \tag{9}$$

where $\xi_t(i)$ represents higher order terms identified in Appendix B.1. We see that the consumption and saving decisions are a function of expected real borrowing rates. However, unlike a textbook model, even to first-order we will see that in equilibrium, the borrowing rate of different households will not necessarily be fully characterized by the monetary policy rate, since the expected return on a given household's portfolio $\tilde{\mu}_t(i) \neq i_t$. Thus, the dynamics of aggregate consumption will depend on the equilibrium determination of all borrowing rates across the term structure.

Importantly, when borrowing rates differ across households, then in equilibrium consumption, labor, and wealth will also differ across households (that is, $C_t(i) \neq C_t(i'), N_t(i) \neq N_t(i'), \mathcal{B}_t(i) \neq \mathcal{B}_t(i')$ for different households $i \neq i'$), which implies that marginal utility is also not equalized.

Intermediate Firms. A continuum of intermediate goods producers index by $j \in [0,1]$ produce differentiated goods $Y_t(j)$ and set prices $\mathcal{P}_t(j)$. Final output Y_t is produced by a competitive retail sector, which aggregates according to $Y_t \equiv \left[\int_0^1 Y_t(j)^{\frac{\epsilon_t - 1}{\epsilon_t}} \, \mathrm{d}j \right]^{\frac{\epsilon_t}{\epsilon_t - 1}}$, where the elasticity of substitution between goods is ϵ_t; exogenous shocks to ϵ_t represent markup shocks (defined below). This implies the follow demand and price index for differentiated goods

$$Y_t(j) = \left(\frac{\mathcal{P}_t(j)}{\mathcal{P}_t} \right)^{-\epsilon_t} Y_t, \quad \mathcal{P}_t = \left[\int_0^1 \mathcal{P}_t(j)^{1-\epsilon_t} \, \mathrm{d}j \right]^{\frac{1}{1-\epsilon_t}}. \tag{10}$$

Each firm j faces the same production function $Y_t(j) = Z_t L_t(j)$, where Z_t is aggregate technology and $L_t(j)$ represents the labor inputs of firm j.

The per-period flow nominal profit of firm j is given by

$$\mathrm{d}\Pi_t(j) = [(1+\tau^y)\mathcal{P}_t(j)Y_t(j) - \mathcal{W}_t L_t(j) - \mathcal{T}_t^y]\,\mathrm{d}t - \mathrm{d}\Theta_t(j)\,, \tag{11}$$

where τ^y is a production subsidy, financed by transfers $\mathcal{T}_t^y = \int_0^1 \tau^y \mathcal{P}_t(j)Y_t(j)\,\mathrm{d}j$. In addition to the wage bill $\mathcal{W}_t L_t(j)$, firms face the following flow costs of production:

$$\mathrm{d}\Theta_t(j) = \frac{\vartheta}{2}\pi_t(j)^2 \mathcal{P}_t Y_t\,\mathrm{d}t + \mathcal{W}_t L_t(j)\,\mathrm{d}\check{R}_t\,. \tag{12}$$

The first term captures nominal rigidities. Firms choose $\pi_t(j)$, the rate of change of their prices ($\mathrm{d}\mathcal{P}_t(j) = \mathcal{P}_t(j)\pi_t(j)\,\mathrm{d}t$). Whenever $\vartheta > 0$, firms face Rotemberg costs of choosing $\pi_t(j) \neq 0$. The second term captures a "working capital" financing friction. The cost of financing is

$$\mathrm{d}\check{R}_t \equiv \int_0^T \beta_t(\tau)\frac{\mathrm{d}P_t^{(\tau)}}{P_t^{(\tau)}}\,\mathrm{d}\tau = \hat{\mu}_t\,\mathrm{d}t + \hat{\boldsymbol{\sigma}}_t\,\mathrm{d}\mathbf{B}_t\,, \tag{13}$$

where $\beta_t \equiv \int_0^T \beta_t(\tau)\,\mathrm{d}\tau$ represents the total fraction of the wage bill which must be financed in advance (taken as given by the firm).[6]

Taking as given CES demand and the costs of production, the firm problem at time $t = 0$ is therefore

$$U_0(j) \equiv \max_{\{\pi_t(j)\}_{t=0}^{\infty}} \mathbb{E}_0 \int_0^{\infty} e^{-\rho t} Q_t^{\mathcal{H}} \frac{\mathrm{d}\Pi_t(j)}{\mathcal{P}_t}. \tag{14}$$

Firms transfer profits to households; we assume that profits are discounted by $e^{-\rho t}Q_t^{\mathcal{H}}$, the average SDF across households.

We focus on a symmetric equilibrium in which each firm chooses the same prices ($\mathcal{P}_t(j) = \mathcal{P}_t, \pi_t(j) = \pi_t$). Then as shown in Appendix B.2, we recover a mostly

[6]For simplicity, we assume that portfolio weights are the same across each firm. Allowing for heterogeneity across firm portfolios as in the household sector would imply additional price dispersion; this would not affect the first-order aggregate dynamics but would have implications for social welfare.

familiar nonlinear Phillips curve given by

$$\mathrm{E}_t \frac{\mathrm{d}\left(Q_t^{\mathcal{H}} Y_t \pi_t\right)}{Q_t^{\mathcal{H}} Y_t} = \left[(\rho - \pi_t)\pi_t + \frac{(\epsilon_t - 1)(1+\tau^y) - \epsilon_t \mathrm{MC}_t}{\vartheta}\right] \mathrm{d}t\,, \tag{15}$$

$$\mathrm{MC}_t \equiv (1+\hat{\mu}_t)\frac{W_t}{Z_t}. \tag{16}$$

Interpreting the exogenous fluctuations in ϵ_t as markup shocks, we see that conditional on marginal costs, the nonlinear Phillips curve in our model is completely standard. The differences from the textbook case come precisely through (16). In particular, our working capital channel depends on the duration-weighted borrowing costs; analogously to the household problem, we will show that even to first-order, in equilibrium $\hat{\mu}_t \neq i_t$, but instead will depend on the entire term structure of borrowing rates.

Arbitrageurs. In addition to the demand for bonds arising from households and firms, bonds are also traded by arbitrageurs. Arbitrageurs actively choose holdings of short- and long-maturity bonds. The representative arbitrageur solves the following mean-variance problem:

$$\max_{\{\mathcal{X}_t(\tau)\}_{\tau=0}^{T}} \mathrm{E}_t\, \mathrm{d}\mathcal{X}_t - \frac{a_t}{2}\mathrm{Var}_t\, \mathrm{d}\mathcal{X}_t \tag{17}$$

$$\text{s.t. } \mathrm{d}\mathcal{X}_t = \mathcal{X}_t i_t\, \mathrm{d}t + \int_0^T \mathcal{X}_t(\tau)\left(\frac{\mathrm{d}P_t^{(\tau)}}{P_t^{(\tau)}} - i_t\, \mathrm{d}t\right)\mathrm{d}\tau\,. \tag{18}$$

That is, they choose nominal bond holdings $\mathcal{X}_t(\tau)$ across all maturities τ. Arbitrageurs choose to engage in carry trades across the term structure in order to optimally satisfy the tradeoff between higher expected returns and the volatility on their balance sheet. The risk-return tradeoff is governed by parameter a_t. Formally this is a risk aversion parameter, but more generally this is a proxy for capital constraints, Value-at-Risk constraints, or any frictions which imply that bond returns are not determined as in a model with a representative household with perfect access to financial securities. We assume that arbitrageurs transfer all gains and losses in their wealth $\mathrm{d}\mathcal{X}_t$ to households each period.

The arbitrageur optimality conditions are given by

$$\mu_t(\tau) - i_t = a_t \int_0^T \mathcal{X}_t(\tau')\boldsymbol{\sigma}_t(\tau)\boldsymbol{\sigma}_t(\tau')^\top \, \mathrm{d}\tau' \,. \tag{19}$$

Hence, arbitrageurs ensure that no risk-free arbitrage opportunities exist. Equation (19) shows that the expected excess return of any τ-maturity bond is a function of the covariance of bond returns across the entire term structure, scaled by arbitrageur risk aversion a_t and weighted by holdings $\mathcal{X}_t(\tau)$ (which in equilibrium will be determined by market clearing).

We assume that $a_t = \frac{a}{\mathcal{P}_t Y}$; that is, risk aversion is constant relative to prices (scaled by steady state output, defined below). Analogous to household wealth, define real arbitrageur bond positions as $X_t(\tau) = \frac{\mathcal{X}_t(\tau)}{\mathcal{P}_t Y}$.

Government. The fiscal authority sets the production subsidy τ^y, which is self-financed through lump-sum taxes on firms. The central bank sets the nominal interest rate i_t, and the fiscal authority pays this interest i_t on short-term bonds (reserves). Besides the production subsidy, the consolidated government may also issue bonds $S_t(\tau) \equiv \frac{\mathcal{S}_t(\tau)}{\mathcal{P}_t Y}$. This represents the issuance of bonds from the fiscal authority net of any asset purchases $\mathrm{QE}_t(\tau)$ from the central bank from QE programs. All gains or losses from the consolidated government's bond issuances are remunerated lump-sum to the households.

As a benchmark, we assume that the central bank can utilize balance sheet tools and the short rate without any constraints. More generally we assume that the central bank is subject to constraints. First, the central bank may face real deadweight costs of adjustment when setting the short-term interest rate:

$$Y_t \Psi_t^i \equiv Y_t \frac{\psi^i}{2} \left(i_t - \bar{i}_t\right)^2 , \tag{20}$$

where $\bar{i}_t$ represents a potentially time-varying (exogenous) policy target. We interpret (20) as capturing in reduced-form constraints such as the effective lower bound; however, to maintain our linear-quadratic approximations, we assume a symmetric loss function. Thus, whenever $\psi^i > 0$, setting the short-term interest rate away from $\bar{i}_t$ will imply deadweight losses.

Second, bond holdings may be subject to real deadweight holding costs (measured

in terms of output) which take the form:

$$Y_t \Psi_t^S \equiv Y_t \int_0^T \frac{\psi(\tau)}{2} \mathrm{QE}_t(\tau)^2 \, \mathrm{d}\tau \,. \tag{21}$$

Thus, whenever $\psi(\tau) > 0$, central bank holdings away from steady state will imply deadweight losses. These costs represent holding costs which the central bank faces (but which do not impact private bond arbitrageurs). These costs can also be thought of as capturing in reduced-form the fiscal costs associated with gains and losses of the central bank balance sheet.[7]

Market clearing. We study a symmetric equilibrium in which all firms make the same decisions and so $Y_t(j) = Y_t$, $\mathcal{P}_t(j) = \mathcal{P}_t$, $\pi_t(j) = \pi_t$, $L_t(j) = L_t$. Define aggregate consumption and labor supply as $C_t = \int_0^1 C_t(i) \, \mathrm{d}i$ and $N_t = \int_0^1 N_t(i) \, \mathrm{d}i$. Market clearing in production, labor markets, and consumption implies

$$Y_t = Z_t L_t = Z_t N_t, \quad C_t = Y_t \left(1 - \frac{\vartheta}{2} \pi_t^2 - \Psi_t^i - \Psi_t^S \right), \tag{22}$$

where aggregate consumption differs from aggregate output due to deadweight loss from price changes when $\vartheta > 0$, or policy frictions when $\psi^i > 0$ or $\psi(\tau) > 0$.

Bond market clearing implies

$$\mathcal{X}_t(\tau) - \beta_t(\tau) \mathcal{W}_t N_t + \int_0^1 \eta_{it}(\tau) \mathcal{B}_t(i) \, \mathrm{d}i = \mathcal{S}_t(\tau) \tag{23}$$

for all maturities $\tau > 0$.

2.2 First- and Second-Order Perturbations

We study perturbations of the nonlinear model as follows. Consider the family of economies indexed by the scalar $\xi \geq 0$, where the diffusion matrix from (1) is parameterized by $\boldsymbol{\sigma}_z(\xi) \equiv \xi \boldsymbol{\sigma}_z$ and arbitrageur risk aversion is parameterized $a(\xi) \equiv \xi^{-2} a$, with $\boldsymbol{\sigma}_z, a$ fixed. Thus we have $a(\xi) \boldsymbol{\sigma}_z(\xi) \boldsymbol{\sigma}_z(\xi)^\top \to a \boldsymbol{\sigma}_z \boldsymbol{\sigma}_z^\top$ as $\xi \to 0$. That is, the product of risk aversion and physical risk remains non-zero and bounded even as risk

[7] If the fiscal authority can only utilize distortionary taxation, then the gains and losses of the central bank balance sheets will induce fiscal costs; see d'Avernas et al. (2024). Finally, we assume that the real costs of these policy frictions are financed via lump-sum taxation of households.

becomes small.[8] The following Proposition characterizes the optimal allocation in the riskless limit.

Proposition 1 (First-best allocation)**.** *Consider the limiting case where $\xi \to 0$ and suppose that wealth is initially equalized across households ($\mathcal{B}_0(i) \equiv \mathcal{B}_0$). Then with perfect arbitrage ($a = 0$), the model is equivalent to a representative agent model. With the optimal production subsidy τ^y given by (A1), the first-best allocation is obtained with flexible prices ($\vartheta = 0$).*

Prop. 1 shows that, so long as the production subsidy is chosen optimally to remove steady-state distortions from imperfect competition and the working capital frictions, there is no room for policy improvement in the flexible price, perfect arbitrage model. Further, in the proof of Prop. 1, we show that if steady-state policy satisfies $i = \rho$ and $S(\tau) = B \int_0^1 \eta_i(\tau)\,\mathrm{d}i - \beta(\tau)$, then the steady state features zero inflation ($\pi = 0$) and zero arbitrageur positions ($X(\tau) = 0$). This in turn implies that steady state in the first-best equilibrium and in the sticky price, imperfect arbitrage equilibrium coincide. In particular, steady-state consumption, labor supply, and wealth across households are equalized ($C(i) = C, N(i) = N, B(i) = B$), and the steady state is efficient.

Log-Linear Perturbation. We study the dynamics of the model around the "low risk, low risk-bearing capacity" limiting case as $\xi \to 0$ but $a \geq 0$. This approximation method allows for tractable first- and second-order approximations of the model, while still allowing for first-order variation in risk premia.[9]

The arbitrageur optimality condition (19) become

$$\mu_t(\tau) - i_t = a\boldsymbol{\sigma}(\tau)\boldsymbol{\Lambda}_t \equiv \lambda_t(\tau), \quad \text{where } \boldsymbol{\Lambda}_t = \int_0^T \boldsymbol{\sigma}(\tau)^\top x_t(\tau)\,\mathrm{d}\tau\,. \tag{24}$$

We see that the expected excess return of any τ-maturity bond is a function of bond-specific risk loadings $\boldsymbol{\sigma}(\tau)$ and a global set of risk prices $\boldsymbol{\Lambda}_t$. The market price of risk depends on arbitrageur risk aversion a as well as holdings $x_t(\tau)$. Note the

[8] This assumption can also be formalized in terms of the size of the intermediary sector relative to the size of asset markets (e.g. see Itskhoki and Mukhin 2023), and captures explicitly that the arbitrageur "risk aversion" parameter a measures the degree of imperfect arbitrage.

[9] For notational convenience, we drop dependence on ξ in the main text. Additionally, for any strictly positive variable, let $A_t \equiv Ae^{a_t}$; for variables which take positive or negative values, let $A_t \equiv A + a_t$.

approximation now implies the bond-specific diffusions $\boldsymbol{\sigma}(\tau)$ are time-invariant, but are still present at first-order. Additionally, these loadings are endogenous.

Denote aggregate output in the first-best equilibrium by Y_t^n, and define the output gap $X_t \equiv \frac{Y_t}{Y_t^n}$. Aggregating (9) across households, the dynamics of the (log) output gap x_t are governed by a modified aggregate Euler equation:

$$\mathrm{E}_t \, \mathrm{d}x_t = \varsigma^{-1} \left[\tilde{\mu}_t - \pi_t - v_t \right] \mathrm{d}t \,. \tag{25}$$

The term v_t is the "natural" rate, which is the real borrowing rate in the first-best allocation from Prop. 1 (and depends on the expected rate of change of technology and discount factor shocks). The "household effective borrowing rate" (in terms of deviation from steady state) is given by

$$\tilde{\mu}_t = \int_0^T \eta(\tau) \mu_t(\tau) \, \mathrm{d}\tau \,, \quad \eta(\tau) \equiv \int_0^1 \eta_i(\tau) \, \mathrm{d}i \,, \tag{26}$$

where $\eta(\tau)$ represents the average across households of the steady-state portfolio weight in τ-maturity bonds.

Re-writing the firm optimality condition in terms of output gaps gives a New Keynesian Phillips curve:

$$\mathrm{E}_t \, \mathrm{d}\pi_t = \left[\rho \pi_t - \kappa \left(x_t + \alpha_{\hat{\mu}} \hat{\mu}_t + \alpha_{\hat{\beta}} \hat{\beta}_t \right) - u_t \right] \mathrm{d}t \,, \tag{27}$$

where κ measures the degree of aggregate rigidity, given by (B2). In addition to the usual drivers of inflation, our model features

$$\hat{\mu}_t = \int_0^T \beta(\tau) \mu_t(\tau) \, \mathrm{d}\tau \,, \quad \hat{\beta}_t = \int_0^T \beta_t(\tau) \, \mathrm{d}\tau \,, \tag{28}$$

which capture the "firm effective borrowing rate" and firm financing shocks (respectively), and the parameters $\alpha_{\hat{\mu}} = \frac{1}{(1+\rho\beta)(\varsigma+\varphi)}, \alpha_{\hat{\beta}} = \frac{\rho}{(1+\rho\beta)(\varsigma+\varphi)}$, where $\beta = \int_0^T \beta(\tau) \, \mathrm{d}\tau$. The markup shock u_t arises from exogenous variation in demand elasticities ϵ_t.

Second-Order Social Welfare Loss. We already see that first-order that aggregate dynamics departure from the textbook model. More importantly, the welfare consequences differ due to the inefficiencies described above. A second-order expansion of social welfare relative to the first-best allocation gives the per-period social

loss

$$\mathcal{L}_t \equiv (\varsigma + \varphi)x_t^2 + \vartheta\pi_t^2 \tag{29}$$

$$+ \frac{\varsigma}{\varphi}(\varphi + \varsigma)\operatorname{Var}_i \check{c}_t(i) \tag{30}$$

$$+ \int_0^T \psi(\tau) q e_t(\tau)^2 \,\mathrm{d}\tau + \psi^i \left(i_t - \bar{i}_t\right)^2 . \tag{31}$$

The first two terms in line (29) capture the welfare losses associated with the nominal rigidities in the model; these terms arise in representative agent New Keynesian (RANK) models. Compared to a standard RANK model, line (30) shows that social welfare loss also depends on the term

$$\operatorname{Var}_i \check{c}_t(i) \equiv \int_0^1 \check{c}_t(i)^2 \,\mathrm{d}i .$$

Thus, increased consumption dispersion across households implies welfare losses, due to imperfect risk-sharing.

The final terms in line (31) represent losses associated with the central bank balance sheet and short rate policies from equations (21) and (20).

Wealth heterogeneity sufficient statistics. The dynamics of the output gap x_t and inflation π_t in (25) and (27) imply that, to a first-order, our model is similar to the representative agent case considered in Ray et al. (2024). However, even to a first-order, our model will feature more complicated dynamics due to the endogenous time-variation in demand for bonds which arises from heterogeneous households. Additionally, we saw that welfare inherently depends on consumption dispersion $\operatorname{Var}_i \check{c}_t(i)$ across households, which itself varies over time. This complexity exists because the continuum of household bond positions is necessary for bond-market clearing, and thus through arbitrageur optimality conditions will affect borrowing rates.

Fortunately, we now show that tracking the entire wealth distribution is not necessary. Instead, only a small number of endogenous sufficient statistics are required. Let $b_t(i)$ denote the real wealth deviations from steady state of household i, and $\check{b}_t(i) \equiv b_t(i) - b_t$ denote the difference from aggregate household wealth $b_t \equiv \int b_t(i)\,\mathrm{d}i$; similarly let $\check{c}_t(i) \equiv c_t(i) - c_t$. Define the following risk-weighted portfolio exposures

$$\boldsymbol{\eta}_i^\sigma \equiv \int_0^T \eta_i(\tau)\boldsymbol{\sigma}(\tau)\,\mathrm{d}\tau\,, \qquad \boldsymbol{\eta}^\sigma \equiv \int_0^1 \boldsymbol{\eta}_i^\sigma\,\mathrm{d}i\,, \qquad \boldsymbol{\beta}^\sigma \equiv \int_0^T \beta(\tau)\boldsymbol{\sigma}(\tau)\,\mathrm{d}\tau\,, \tag{32}$$

so that integrating (24) against the portfolio weights gives

$$\tilde{\mu}_t(i) = i_t + a\boldsymbol{\eta}_i^\sigma \boldsymbol{\Lambda}_t, \qquad \tilde{\mu}_t \equiv \int_0^1 \tilde{\mu}_t(i)\,\mathrm{d}i = i_t + a\boldsymbol{\eta}^\sigma \boldsymbol{\Lambda}_t, \qquad \hat{\mu}_t = \beta i_t + a\boldsymbol{\beta}^\sigma \boldsymbol{\Lambda}_t. \tag{33}$$

Then letting f_t and $\boldsymbol{\sigma}_f$ denote the drift and diffusion of aggregate net non-labor transfer flow to households, aggregate wealth evolves according to

$$\mathrm{d}b_t = \left[\rho b_t + B\left(\tilde{\mu}_t - \pi_t\right) + (\varsigma + \varphi)x_t + f_t\right]\mathrm{d}t + \left[B\boldsymbol{\eta}^\sigma + \boldsymbol{\sigma}_f\right]\mathrm{d}\mathbf{B}_t\,. \tag{34}$$

Note that with zero steady state wealth or bond supply, $\boldsymbol{\sigma}_f = \mathbf{0}$ and $f_t = -(\varsigma + \varphi)x_t$, so that $b_t \equiv 0$; but with $B \neq 0$, revaluation effects imply aggregate wealth is a non-trivial state variable.

Additionally, relative wealth of household i evolves according to

$$\mathrm{d}\check{b}_t(i) = \left[\rho\check{b}_t(i) + B\check{\lambda}_t(i) - (1 + \varsigma/\varphi)\,\check{c}_t(i)\right]\mathrm{d}t + B\left(\boldsymbol{\eta}_i^\sigma - \boldsymbol{\eta}^\sigma\right)\mathrm{d}\mathbf{B}_t\,, \tag{35}$$

where $\check{\lambda}_t(i) \equiv \tilde{\mu}_t(i) - \tilde{\mu}_t = a(\boldsymbol{\eta}_i^\sigma - \boldsymbol{\eta}^\sigma)\boldsymbol{\Lambda}_t$ is the relative portfolio premium and differential consumption $\check{c}_t(i)$ evolves according to

$$\mathrm{E}_t\,\mathrm{d}\check{c}_t(i) = \varsigma^{-1}\check{\lambda}_t(i)\,\mathrm{d}t = \varsigma^{-1}a\left(\boldsymbol{\eta}_i^\sigma - \boldsymbol{\eta}^\sigma\right)\boldsymbol{\Lambda}_t\,\mathrm{d}t\,. \tag{36}$$

Unlike aggregate wealth, even with $B = 0$, relative wealth is generically non-zero. However, heterogeneity is entirely driven by risk exposure differences $\boldsymbol{\eta}_i^\sigma - \boldsymbol{\eta}^\sigma$, which is an N_z-dimensional vector. This insight motivates the following risk-weighted cross-

sectional aggregates:

$$\mathbf{b}_t^\sigma \equiv \int_0^1 (\boldsymbol{\eta}_i^\sigma)^\top \check{b}_t(i)\,\mathrm{d}i\,, \quad \check{\mathbf{c}}_t^\sigma \equiv \int_0^1 (\boldsymbol{\eta}_i^\sigma)^\top \check{c}_t(i)\,\mathrm{d}i\,, \tag{37}$$

and the positive semi-definite matrix $\check{\boldsymbol{\Sigma}} \equiv \int_0^1 (\boldsymbol{\eta}_i^\sigma - \boldsymbol{\eta}^\sigma)^\top (\boldsymbol{\eta}_i^\sigma - \boldsymbol{\eta}^\sigma)\,\mathrm{d}i$. The following Theorem shows that these objects are sufficient for characterizing the aggregate first-order dynamics of the model, risk prices, and the time-variation in cross-sectional consumption dispersion.

Theorem 1 (Sufficient statistics)**.** *Suppose initial relative wealth is spanned by exposures: $\check{b}_0(i) = (\boldsymbol{\eta}_i^\sigma - \boldsymbol{\eta}^\sigma)\boldsymbol{\gamma}_0^b$ for some $\boldsymbol{\gamma}_0^b \in \mathbb{R}^{N_z}$. Then:*

1. *For all t, $\check{b}_t(i) = (\boldsymbol{\eta}_i^\sigma - \boldsymbol{\eta}^\sigma)\boldsymbol{\gamma}_t^b$ and $\check{c}_t(i) = (\boldsymbol{\eta}_i^\sigma - \boldsymbol{\eta}^\sigma)\boldsymbol{\gamma}_t^c$, where*

$$\mathrm{d}\boldsymbol{\gamma}_t^b = \left[\rho\boldsymbol{\gamma}_t^b + aB\boldsymbol{\Lambda}_t - (1+\varsigma/\varphi)\boldsymbol{\gamma}_t^c\right]\mathrm{d}t + B\,\mathrm{d}\mathbf{B}_t\,, \quad \mathrm{E}_t\,\mathrm{d}\boldsymbol{\gamma}_t^c = a\varsigma^{-1}\boldsymbol{\Lambda}_t\,\mathrm{d}t\,. \tag{38}$$

2. *The objects (37) satisfy $\mathbf{b}_t^\sigma = \check{\boldsymbol{\Sigma}}\boldsymbol{\gamma}_t^b$ and $\check{\mathbf{c}}_t^\sigma = \check{\boldsymbol{\Sigma}}\boldsymbol{\gamma}_t^c$.*

3. *Cross-sectional consumption dispersion is given by*

$$\mathrm{Var}_i\,\check{c}_t(i) = \boldsymbol{\gamma}_t^{c\top}\check{\boldsymbol{\Sigma}}\boldsymbol{\gamma}_t^c = \check{\mathbf{c}}_t^{\sigma\top}\check{\boldsymbol{\Sigma}}^+\check{\mathbf{c}}_t^\sigma. \tag{39}$$

4. *Risk prices are given by*

$$\boldsymbol{\Lambda}_t = (\boldsymbol{\beta}^\sigma)^\top\left[(1+\varsigma+\varphi)x_t + \frac{1+\varphi}{\varsigma+\varphi}z_t\right] + \mathbf{s}_t^\sigma - \mathbf{q}\mathbf{e}_t^\sigma - (\boldsymbol{\eta}^\sigma)^\top b_t - \mathbf{b}_t^\sigma, \tag{40}$$

   where the balance sheet and supply terms are defined $\mathbf{q}\mathbf{e}_t^\sigma \equiv \int_0^T \boldsymbol{\sigma}(\tau)^\top q e_t(\tau)\,\mathrm{d}\tau$, $\mathbf{s}_t^\sigma \equiv \int_0^T \boldsymbol{\sigma}(\tau)^\top s_t(\tau)\,\mathrm{d}\tau$.

Thus, since $\boldsymbol{\Lambda}_t$ depends on the household side only through $\mathbf{b}_t^\sigma$ and aggregate wealth b_t, the entire cross-section is summarized by $(\mathbf{b}_t^\sigma, \check{\mathbf{c}}_t^\sigma)$ together with the scalar b_t.

The key insight of Theorem 1 is that what matters to arbitrageurs is risk exposure, not the positions taken in specific bonds *per se*. This is a function of the sources of aggregate volatility in the model, which inherently is low-dimensional. Then because differential consumption choices arise precisely because borrowing rates differ across households; but the same insight shows that differential borrowing rates arise due to

differential risk exposure. Thus all relevant first- and second-order properties of the model can be summarized in this small set of risk-weighted wealth statistics. However, it is important to note that, because bond volatility loadings are endogenous, all of the household and firm risk-weighted objects are themselves endogenous as well.[10]

2.3 Equilibrium

Intuitively, what does general equilibrium look like in this model? From the perspective of households, the key factor is how sensitive their borrowing rates are to the short rate (and movements in asset demand, including balance sheet policies). The model reduces to a benchmark New Keynesian model when these rates move one-for-one, but in general $\mu_t(\tau) \neq i_t$. Suppose that long-term borrowing rates are highly responsive to the policy rate. Then household borrowing is also highly sensitive to the policy rate, and therefore the growth rate of consumption will also react strongly to the policy rate. On the other hand, when long-term rates are insensitive to the policy rate, the pass-through of changes in the policy rate to households is weakened. Through the borrowing decisions of the household, the growth rate of consumption is less responsive to the policy rate.

However, the sensitivity of the effective borrowing rate to the policy rate is an equilibrium object, which also depends on financial markets. Bond prices will adjust in order to achieve equilibrium in bond markets, such that arbitrageurs' portfolio allocation satisfies their mean-variance tradeoff while also clearing the market given the demand from households and firms. In this model, arbitrage is imperfect and the term structure will not be characterized by the expectations hypothesis except under special circumstances. Therefore, it is the risk-adjusted dynamics of the macroeconomy which determine bond prices in financial markets, rather than the actual dynamics of the short rate only.

In general, the term structure will be determined by complicated interactions between the different classes of investors in bond markets (arbitrageurs, firms, and households). Because differentiated households also make consumption and savings decisions as a function of different borrowing rates, the general equilibrium dynamics of the macroeconomy will also depend on these interactions. General equilibrium is obtained when these two forces balance. Thus, characterizing equilibrium involves two

[10]Note that the spanning assumption we make use of in Theorem 1 is relatively innocuous, as the steady state features no wealth dispersion and therefore is trivially satisfied.

key steps: first, understanding the differences between the actual and risk-adjusted dynamics of the economy; and second, linking household savings and consumption choices with the bond prices determined in imperfect financial markets.

We first make use of the following Proposition to characterizing the behavior of asset prices, conditional on some arbitrary set of state variables.

Proposition 2 (Asset prices)**.** *Suppose that the first-order equilibrium admits a stationary linear representation with an N_s-dimensional state variable $\mathbf{s}_t$, which satisfies $i_t = \boldsymbol{\iota}^\top \mathbf{s}_t$ and $\boldsymbol{\Lambda}_t = \mathbf{L}\mathbf{s}_t$, and evolves according to*

$$\mathrm{d}\mathbf{s}_t = -\boldsymbol{\Gamma}\mathbf{s}_t \,\mathrm{d}t + \boldsymbol{\sigma}_s \,\mathrm{d}\mathbf{B}_t \,. \tag{41}$$

Then an affine (log) bond price equilibrium with

$$-\log P_t^{(\tau)} = \mathbf{A}(\tau)^\top \mathbf{s}_t + C(\tau), \tag{42}$$

implies that the affine coefficients are given by

$$\mathbf{A}(\tau) = \left[\mathbf{I} - e^{-\mathbf{M}\tau}\right] \mathbf{M}^{-1}\boldsymbol{\iota}, \quad \mathbf{M}^\top = \boldsymbol{\Gamma} + a\boldsymbol{\sigma}_s\mathbf{L}. \tag{43}$$

The matrix $\mathbf{M}$ can be thought of as the risk-adjusted dynamics of the state. In the first-best, arbitrageurs are perfectly risk-neutral ($a = 0$), so we have $\mathbf{M} = \boldsymbol{\Gamma}^\top$. However, when $a \neq 0$, $\mathbf{M}$ additionally depends on risk prices (through $\mathbf{L}$); and risk prices will themselves depend on $\mathbf{M}$. Thus, in equilibrium, $\mathbf{M}$ appears on both sides of equation (43).

Note the similarity with Vayanos and Vila (2021). Conditional on the equilibrium dynamics of the model, the only difference for bond pricing comes from the more general formulation of the dynamics of the short-term interest rate and from the market clearing conditions in bond markets.

Prop. 2 characterizes risk pricing, taking as given aggregate dynamics. We now approach the problem from the other direction.

Proposition 3 (Aggregate dynamics)**.** *Collect all risk factors $\mathbf{z}_t$ as well as aggregate wealth b_t and the wealth statistics $\mathbf{b}_t^\sigma$ defined in (37) into the vector $\mathbf{s}_t$. Collect the output gap x_t, inflation π_t, and the consumption statistics $\check{\mathbf{c}}_t^\sigma$ into the vector $\mathbf{y}_t$. Assume that the short rate i_t, bond supply $s_t(\tau)$, and balance sheet tools $qe_t(\tau)$ are functions of these objects.*

Then given $\check{\boldsymbol{\Sigma}}$, $\boldsymbol{\eta}^{\sigma}$, *and* $\boldsymbol{\beta}^{\sigma}$ *defined in* (32), *the linear rational expectations equilibrium is given by*

$$\mathrm{d}\mathbf{s}_t = -\boldsymbol{\Gamma}\mathbf{s}_t\,\mathrm{d}t + \boldsymbol{\sigma}_s\,\mathrm{d}\mathbf{B}_t\,, \quad \mathbf{y}_t = \boldsymbol{\Omega}\mathbf{s}_t, \tag{44}$$

where the matrices $\boldsymbol{\Gamma}, \boldsymbol{\Omega}$, *and* $\boldsymbol{\sigma}_s$ *are given in Appendix B.3.*

With the results in Propositions 2 and 3, we can characterize the equilibrium of the model.

Theorem 2 (Existence and uniqueness)**.** *An affine equilibrium is one in which the state and jump variables evolve according to equations* (44), *and asset prices are determined by the solution to the expressions* (43). *The per-period loss function can be written* $\mathcal{L}_t = \mathbf{s}_t^{\top}\Psi\mathbf{s}_t$. *In a neighborhood of risk-neutrality (*$a \approx 0$*), the equilibrium exists and is (locally) unique.*

Note that the dynamics matrix of the state $\boldsymbol{\Gamma}$ depends on the affine coefficients $\mathbf{A}(\tau)$ through the effective borrowing rates of the households, which itself is a function of the risk-adjusted dynamics matrix $\mathbf{M}$. Thus, equilibrium is determined as a fixed point that produces asset price dynamics consistent with equilibrium dynamics of the macroeconomy and vice versa. In general, an affine equilibrium of this type may not exist, or there may be multiple solutions to this fixed point problem. However, when $a = 0$, the model reduces to a standard New Keynesian model. The result in Theorem 2 shows that this equilibrium persists and is locally unique as we depart from risk neutrality.

Although Theorem 2 characterizes the first-order dynamics and second-order welfare properties of the model, in general it provides little guidance on how either differ from textbook models. In the next section, we focus on simplified versions of the model to derive clear predictions regarding dynamics.

3 Aggregate Dynamics: Analytical Results

Before studying welfare consequences, we explore the first-order dynamics of the model. To do so, we consider versions of the model where the central bank follows ad-hoc policy rules.

We focus on a version of the model with only one source of aggregate uncertainty: natural rate shocks $v_t \equiv -\varsigma\kappa_z \frac{1+\varphi}{\varsigma+\varphi} z_t$, which are driven by technology z_t only. We simplify further for most of this section and study the case of fully rigid prices ($\vartheta \to \infty$), and where household wealth is zero in steady state ($B = 0$). In this case, technology follows a simple exogenous Ornstein-Uhlenbeck process with mean reversion denoted by κ_z and diffusion denoted by σ_z; the dynamics of the rest of the economy are summarized by the following system of equations

$$\mathrm{d}b_t^\sigma = \left[-\left(1 + \frac{\varsigma}{\varphi}\right) \check{c}_t^\sigma + \rho b_t^\sigma \right] \mathrm{d}t \,, \tag{45}$$

$$\mathrm{E}_t \, \mathrm{d}x_t = \varsigma^{-1} \left[i_t + a\eta^\sigma \lambda_t - v_t \right] \mathrm{d}t \,, \tag{46}$$

$$\mathrm{E}_t \, \mathrm{d}\check{c}_t^\sigma = \varsigma^{-1} a \check{\Sigma} \lambda_t \, \mathrm{d}t \,, \tag{47}$$

$$\lambda_t = -q e_t^\sigma + \beta^\sigma \left((1 + \varsigma + \varphi) x_t - \frac{1}{\varsigma\kappa_z} v_t \right) - b_t^\sigma . \tag{48}$$

With one aggregate risk factor, note that risk prices λ_t and the other composite endogenous risk objects are now scalars. Note that in this version of the model, the financing frictions which firms face does not impact their pricing decision (as by assumption, prices are fully rigid). However, through market clearing, firm demand for bonds implies that aggregate fluctuations lead to fluctuations in risk premia through market clearing.

Throughout, we assume that the central bank follows Taylor-type of rules for its policy tools given by $i_t = \phi_x x_t + \epsilon_{i,t}$, $q e_t^\sigma = \phi_x^\sigma x_t + \epsilon_{q,t}^\sigma$ (note that for dynamics, we only need to specify the risk-weighted path of the central bank's balance sheet). The terms $\epsilon_{i,t}$ and $\epsilon_{q,t}^\sigma$ represent zero-probability monetary policy shocks, which evolve deterministically according to

$$\mathrm{d}\epsilon_{i,t} = -\kappa_i \epsilon_{i,t} \, \mathrm{d}t \,, \quad \mathrm{d}\epsilon_{q,t}^\sigma = -\kappa_q \epsilon_{q,t}^\sigma \, \mathrm{d}t \,.$$

We first consider a benchmark case where there is no working capital friction and so firms do not borrow in bond markets, and the central bank does not systematically use balance sheet tools. The following Proposition derives results regarding dynamics for conventional and unconventional monetary policy.

Proposition 4 (Rigid price dynamics, no firm borrowing)**.** *Suppose that $\beta(\tau) = 0$ and $\phi_x^\sigma = 0$. If household wealth is initially equalized at steady state (so $b_t^\sigma = 0$), then*

the equilibrium dynamics of the model are equivalent to a representative agent model, and $\lambda_t = 0$. *We have*

$$\frac{\partial x_t}{\partial v_t} > 0, \quad \frac{\partial \lambda_t}{\partial v_t} = 0, \quad \frac{\partial x_t}{\partial \epsilon_{i,t}} < 0, \quad \frac{\partial \lambda_t}{\partial \epsilon_{i,t}} = 0.$$

However, following an unconventional monetary policy shock, we have

$$\frac{\partial x_t}{\partial \epsilon^{\sigma}_{q,t}} > 0, \quad \frac{\partial \lambda_t}{\partial \epsilon^{\sigma}_{q,t}} < 0,$$

and the model is no longer equivalent to a representative agent model, as $b^{\sigma}_{t+k} \neq 0$ *following the shock.*

Prop. 4 shows that without firm borrowing and initial equalization of asset positions across households, the model which only features conventional monetary policy is identical to a representative agent model. If household wealth is initially at steady state, then arbitrageurs initially take zero positions in bond markets and hence do not require any risk compensation. Following movements in the short-term interest rate (either endogenously or due to policy shocks), while the macroeconomy moves away from steady state, household wealth does not change. Without firm borrowing, this implies that arbitrageurs continue to maintain zero positions, and thus risk prices do not move. Thus, the dynamics are equivalent to a representative agent model.

However, even without firm borrowing and with an equalized wealth distribution, the predictions of the model differ from that of a representative agent model when considering unconventional policy. First, because an increase in asset purchases by the central bank reduces the holdings of arbitrageurs on impact, required risk compensation falls. Thus, holding fixed the short-term interest rate, the "effective" borrowing rate faced by households falls as well, which is stimulative. But the decline in borrowing rates is larger for households who borrow with long-maturity assets. Thus, the induced increase in consumption is larger for these households than for households with access to short-maturity bonds. Thus, consumption dispersion $\check{c}^{\sigma}_t$ becomes non-zero, and the risk-weighted asset positions of households b^{σ}_t begin to move away from steady state.

Prop. 4 shows that conventional and unconventional policies may be substitutes in the sense that shocks to either tool can boost output, but the transmission mechanisms and implied dynamics following the shocks differ in meaningful ways. The

welfare implications will therefore differ as well (which we return to in later sections).

The next Proposition shows that when firms also borrow as part of their production process, the dynamics of the model differ more significantly from a representative agent model.

Proposition 5 (Rigid price dynamics, general case)**.** *Suppose that $\beta^\sigma > 0$. If $0 < a < \bar{a}$ for some upper bound on arbitrageur risk aversion, and the Taylor rule coefficient satisfies $0 < \phi_x < \bar{\phi}_x \equiv \varsigma\kappa_z(\varsigma + \varphi)$, then following a technology shock which decreases the output gap, both the short-term rate i_t and term premia λ_t decline on impact. However, for some $k > \bar{k}$, term premia λ_{t+k} eventually turn positive, while the short rate remains below steady state. These effects are larger when the central bank also utilizes balance sheet tools ($\phi_x^\sigma < 0$).*

The intuition for the results of Prop. 5 can be seen when thinking about the dynamics of the model near risk neutrality. Under risk neutrality, we have that the output gap is an increasing function of natural rates: $x_t \propto v_t$. When $\phi_x < \bar{\phi}_x$, the reaction of firm borrowing in a recession caused by a decline in the natural rate is to reduce borrowing. When $\beta^\sigma > 0$, this implies that λ_t declines. However, the decline in term premia induces households with long-duration investments to reduce savings (or increase borrowing). This must be funded by arbitrageurs, and eventually implies that arbitrageurs are taking long positions in the bond carry trade, and that these positions are larger for longer maturities.

Thus, our model predicts that unconditionally, decreases in the short-term interest rate are associated with decreases in term premia over short-horizons; but over longer horizons, the reaction of term premia becomes insubstantial or even negative. Moreover, when the central bank utilizes balance sheet tools, the unconditional reaction of term premia over short horizons is larger.

It is important to note, however, that this prediction is an unconditional one: aggregate shocks which induce the central bank to ease policy are associated with initial declines in term premia, but over longer horizons this effect flips. But our model makes different predictions for the *conditional* reaction of term premia to monetary policy *shocks*, as shown in the following Corollary.

Corollary 5.1 (Rigid price dynamics, monetary shocks)**.** *Following a conventional monetary policy shock $\epsilon_{i,t}$ which decreases the policy rate i_t, the output gap x_t increases. The conditional reaction of λ_t to i_t is negative, and smaller than the uncon-*

ditional reaction in Prop. 5 for similarly sized movements in the short rate (assuming ϕ_x is not too large).

The difference between the unconditional and conditional reaction of term premia to movements in the short rate arise because of the endogenous movements of quantities in asset markets. In this version of the model, all else equal, firm borrowing is positively correlated with the output gap. However, technology-driven declines in the output gap also induce additional upward pressure on firm borrowing. A reduction in firm borrowing implies a decline in risk compensation required by arbitrageurs. Simultaneously, the short-term interest rate declines due to the endogenous reaction of the policy rule. Because in this one risk-factor model, all stochastic dynamics of the model are due to natural rate (technology) shocks, the unconditional correlation between the policy rate and term premia is therefore positive whenever the Taylor rule is not overly reactive. However, because an unexpected decline in the short-term interest rate is expansionary, the correlation between term premia and the short-term interest rate *conditional on a monetary policy shock* is negative.

Further, the endogenous dynamics of the model induced by slow-moving changes in the relative wealth distribution across households implies that these predictions change when looking over longer horizons. Finally, the quantitative predictions change in version of the model where the central bank is also pursuing balance sheet policies.

4 Empirical Evidence

In order to test our predictions from Section 3, we utilize a similar empirical methodology to Hanson et al. (2021). Rather than attempting to decompose the observed movements in the yield curve into expectation and term premia components (which involves imposing additional structure on the observable data), we analyze the dynamics of the term structure of forward rates. After documenting a number of stylized facts about the dynamics of forward rates, in Section 4.3 we relate these to our model predictions.

We run regressions of the form

$$f_{t+h}^{(\tau)} - f_{t-1}^{(\tau)} = \alpha(\tau) + \beta(\tau) D_t + \epsilon_t(\tau), \tag{49}$$

where $f_t^{(\tau)}$ are τ-year ahead forward rates. Our time frequency is daily, and so $f_{t+h}^{(\tau)} -$

$f_{t-1}^{(\tau)}$ is the h-day-ahead horizon change in forward rates on date t. We explore different choices of right-hand-side variable D_t, as well as maturity τ and horizon h.

Our first set of *unconditional* regressions sets the right-hand-side variable as $D_t = y_t^{(1)} - y_{t-1}^{(1)}$, where $y_t^{(1)}$ is the 1-year spot yield on day t. Thus, our right-hand-size variable is the one-day change in short-term (1-year) yields. We estimate (49) on all days in our sample.[11]

The second set of *conditional* regressions uses high-frequency measures of monetary policy shocks in place of D_t. Our baseline uses the shock series from Nakamura and Steinsson (2018), but results are similar for other choices. We compare the estimate for $h = 0$ and $h > 0$ (our baseline uses 90-day-ahead regressions, but results are very similar for other longer horizon choices). We use the daily nominal forward and spot yields from Gürkaynak et al. (2007) in all our regressions.[12] We estimate (49) across the entire sample as well as over rolling 5-year windows.

4.1 Forwards and Spot Yields, Unconditional

Figure 1 reports our unconditional results using daily changes in the 1-year spot yield. We report the unconditional estimates of (49) over the entire term structure; the x-axis ranges over maturities $\tau = 2$ to $\tau = 20$ years, and hence each point reports a different estimated slope coefficient. The line labeled "short-horizon" reports estimates using one-day ($h = 0$) changes in forward rates, while the "long-horizon" estimates are for three-month ($h = 90$) changes.

First, we see that over short horizons, the slope coefficient is economically large and significant across the entire term structure. Even for very long-maturity forward rates, a 100 basis point increase in the 1-year spot yield is associated with over 40 basis point increase in the $\tau = 20$-year forward rate.

However, the pattern changes significantly when moving towards longer-horizon

[11] Our sample begins in 1982, the date at which high-quality estimates of longer maturity spot and forward yields (up to 20 years) are available. We end the sample in 2020 to avoid including the COVID period in our estimates, although results are similar if we include the COVID period.

[12] We use nominal yield curve data because high-quality estimates are available for both short- and long-maturity spot and forward yields starting since the early 1980s (and even earlier for shorter maturities). In contrast, Treasury Inflation-Protected Securities (TIPS) were only introduced in the late 1990s, and have significantly lower trading volume than nominal Treasury securities; high-quality real yield curve estimates are at best available starting in the mid-2000s, but this creates further issues as the majority of the estimation sample will take place during the zero lower bound period.

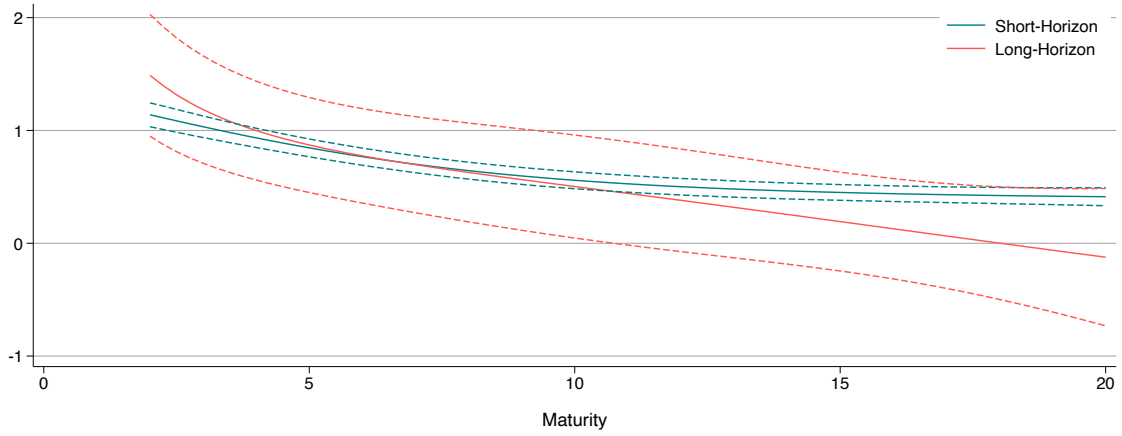


Figure 1: Forward Rates (Unconditional, Varying Maturities)

Notes: estimates from (49) across the term structure; unconditional specification. The estimates labeled short-horizon use $h = 0$ (one day changes), while the estimates labeled long-horizon use $h = 90$ (three month changes). The x-axis ranges over maturities $\tau = 2$ to $\tau = 20$ years. 95% Newey-West confidence intervals are included.

differences. We still find that the daily change in 1-year spot yields is associated with large increases in short-horizon forward rates even three months later (if anything, the point estimate is larger for short maturities when using three-month-ahead changes); but the associated change in long-maturity forward rates ($\tau = 10$ or greater) becomes statistically insignificant, and the point estimates become smaller. For very long maturity forwards, the point estimate becomes negative (though statistically indistinguishable from zero).

Figure 2 studies the behavior of 2-year and 20-year forward rates (the estimates labeled short-maturity and long-maturity, respectively) in more detail across different horizons. The x-axis varies from $h = 0$ to $h = 150$, which corresponds to estimates of (49) using 1-day changes to roughly 5-month changes. We see that unconditionally, a daily increase in 1-year spot yields is associated with a roughly one-to-one increase in 2-year forward rates over all horizons. However, 20-year forward rates respond significantly only in the first month, and point estimates become negative after about two months (though not significantly so).

Figures 3 and 4 repeat the unconditional regressions for $\tau = 20$-year forwards, but over rolling (5-year) windows. Figure 3 uses short-horizon ($h = 0$) changes, while Figure 4 uses long-horizon ($h = 90$) changes. Consistent with our findings over the

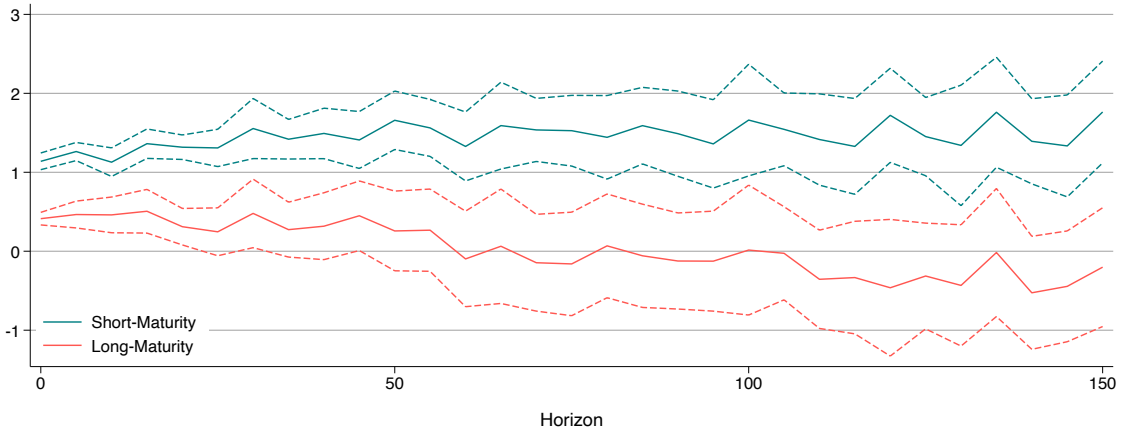


Figure 2: Forward Rates (Unconditional, Varying Horizon)

Notes: estimates from (49) across the different horizons; unconditional specification. The estimates labeled short-maturity use $\tau = 2$ (2-year forwards), while the estimates labeled long-maturity use $\tau = 20$ (20-year forwards). The x-axis ranges over horizons $h = 0$ to $h = 150$ (1-day to 5 months). 95% Newey-West confidence intervals are included.

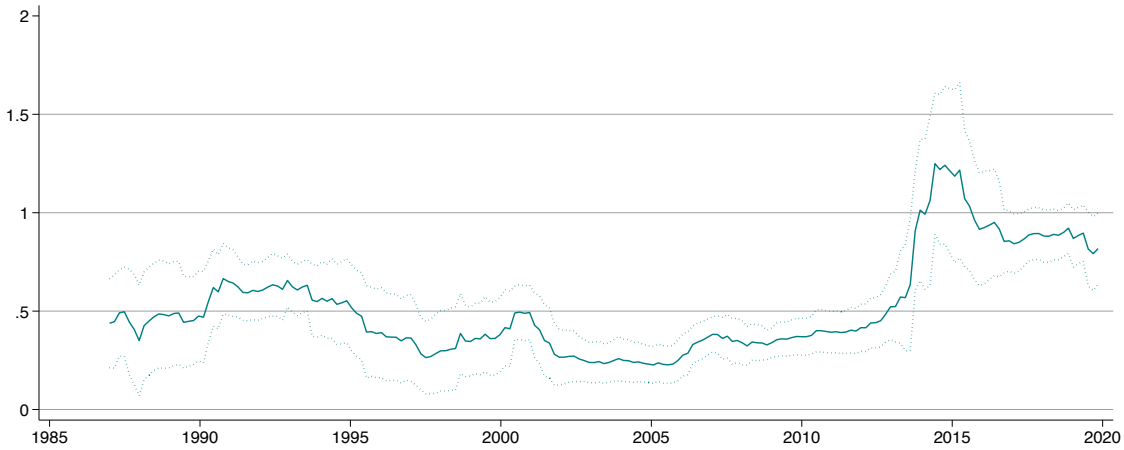


Figure 3: 20-year Forwards (Unconditional, Rolling Short Horizon)

Notes: rolling estimates from (49) for $\tau = 20$-year forward rates; unconditional specification, using short-horizon ($h = 0$) changes. The rolling window is five years; the x-axis includes the end date of each rolling estimate. 95% Newey-West confidence intervals are included.

full sample, we see that over short horizons, unconditionally forward rates move in the same direction as short-term yields, and the co-movement between the two is strong. In addition, the time-variation in our estimates observed in Figure 3 also

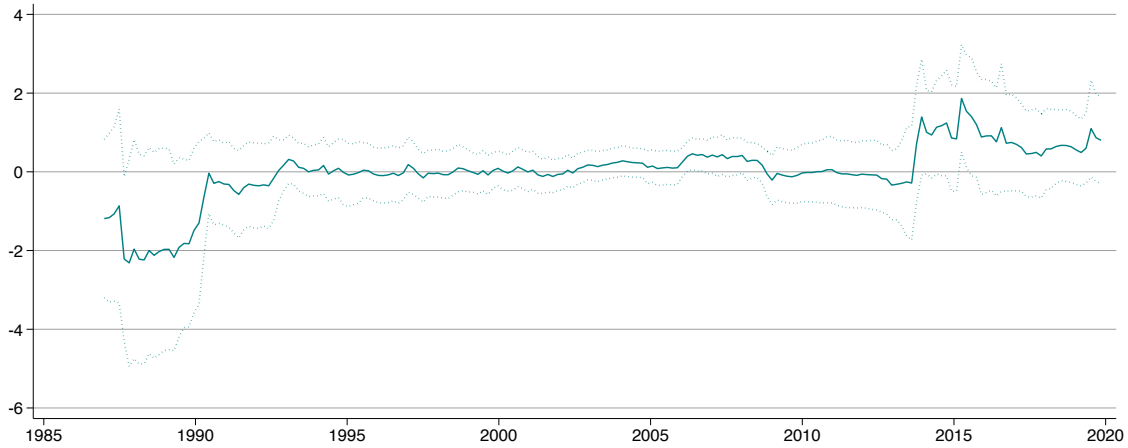


Figure 4: 20-year Forwards (Unconditional, Rolling Long Horizon)

Notes: rolling estimates from (49) for $\tau = 20$-year forward rates; unconditional specification, using long-horizon ($h = 90$) changes. The rolling window is five years; the x-axis includes the end date of each rolling estimate. 95% Newey-West confidence intervals are included.

reveals interesting properties. The unconditional relationship between long-maturity forward rates and short-maturity spot yields becomes the strongest in the end of the sample. The point estimates jump sharply when the sample includes the years 2009-2014, and even become greater than one (though the confidence intervals include values less than one as well).

Figure 4 confirms that over all different sub-samples, long horizon movements in forward rates are much less tied to short-horizon movements in short rates. The estimates are virtually always statistically insignificant, and besides the sample period starting in 2009, the point estimates are economically small and frequently turn negative as well.

4.2 Forwards and Spot Yields, Monetary Shocks

We now turn to our conditional regressions, where we estimate (49) using identified monetary policy shocks. Figure 5 reports our estimates over the entire sample using short-horizon ($h = 0$) changes.[13] Under the assumption that the high-frequency

[13]Results are similar but significantly more noisy for long-horizon regressions. Note that because the conditional specification requires identified monetary shocks, we have significantly fewer observations than the unconditional specification.

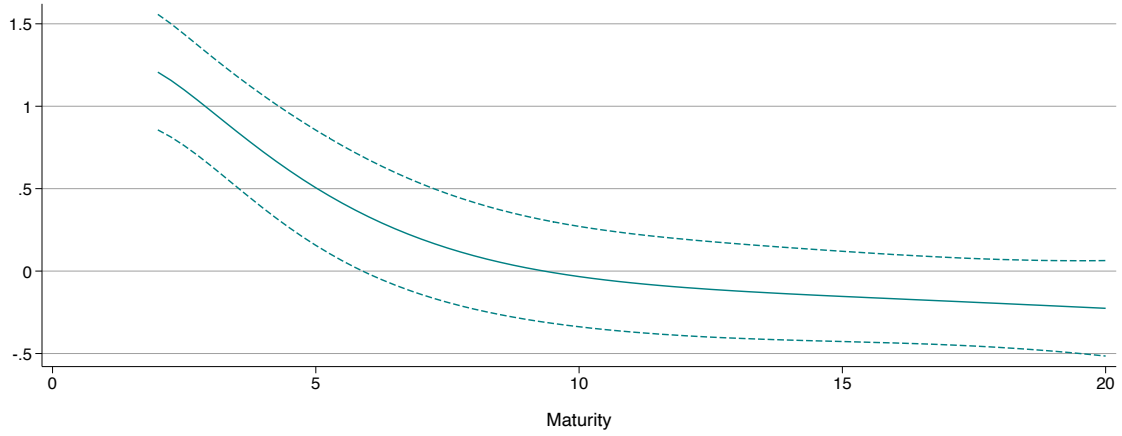


Figure 5: Forwards Rates (Shocks, Varying Maturity)

Notes: estimates from (49) across the term structure; conditional monetary shock specification. We report estimates using $h = 0$ (one day changes) across the term structure. The x-axis ranges over maturities $\tau = 2$ to $\tau = 20$ years. 95% Newey-West confidence intervals are included.

approach to identifying monetary shocks is valid, we can interpret the estimates in this section causally. We see that a contractionary monetary policy shock causes an increase in short-maturity forward rates on impact. However, in contrast with our unconditional findings, the transmission of monetary shocks to intermediate maturity forwards is insignificant (for forward rates with maturity of roughly 5 years onward). Moreover, the point estimate becomes negative for 10-year forward rates, and continues to decline as we move further out in the term structure. The point estimate for $\tau = 20$-year maturities is roughly -0.25, and significantly different from zero at the 90% level (though not at the 95% level).

Finally, Figure 6 repeats the analysis from Figure 5, but across rolling windows. We focus on long-maturity ($\tau = 20$-year) forward rates. Across all sub-samples, the point estimate is virtually always statistically insignificant. Additionally, point estimate are economically small, and if anything are typically negative (although using rolling regressions with identified monetary shocks implies that each sub-sample has only a modest number of observations).

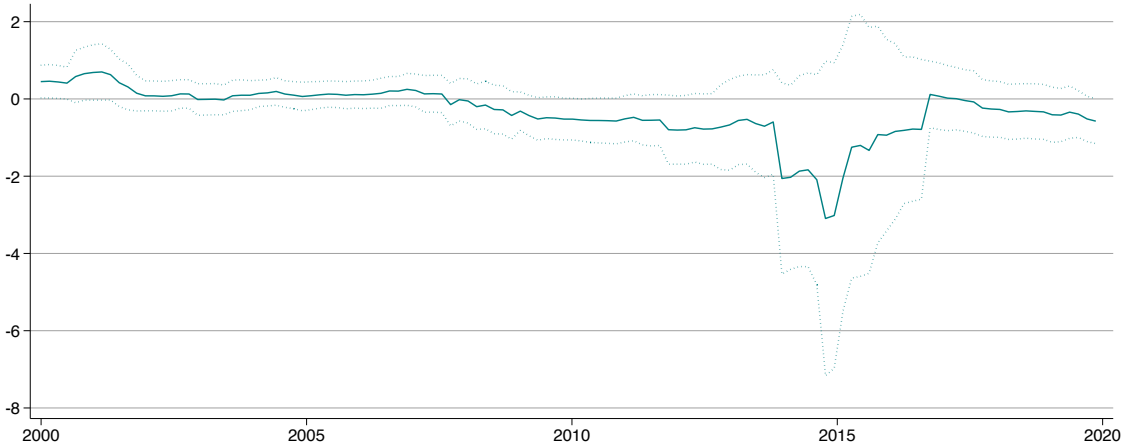


Figure 6: 20-year Forwards (Shocks, Rolling)

Notes: rolling estimates from (49) for $\tau = 20$-year forward rates; conditional monetary shock specification, using short-horizon ($h = 0$) changes. The rolling window is five years; the x-axis includes the end date of each rolling estimate. 95% Newey-West confidence intervals are included.

4.3 Comparison with Model Predictions

In order to link our model-based predictions with our empirical findings, we need to translate our results regarding forward rates to those regarding term premia. Rather than attempting to extract components corresponding to expected path and risk compensation, we take a more qualitative approach. Under the assumption that market participants expect the daily changes in short-term rates to have largely reverted to steady state after 10 to 20 years, the movement in long-maturity forward rates mostly reflect fluctuations in term premia. With this interpretation in mind, we discuss our empirical results and how they relate to the predictions of the model.

First, the results regarding short-horizon movements in forward rates in Figure 1 confirms a key prediction of our model: over short horizons, term premia and short rates move together unconditionally. Next, the time-variation in our estimates observed in Figure 3 is consistent with another one of our predictions. During periods when the central bank utilizes balance sheet tools more than the short rate, the unconditional relationship between term premia and short rates becomes stronger. Third, the results regarding longer-horizon movements in forward rates in Figures 1, 2, and 4 confirms another prediction of our model: term premia movements over longer horizons are less reactive to short-horizon movements in short rates unconditionally, and

the reaction may turn negative. Finally, Figures 5 and 6 confirm another important and sharp prediction of our model: *conditional on monetary shocks*, the reaction of forward rates is weak, and if anything implies that term premia move in the opposite direction of monetary policy *shocks*.

The stark qualitative difference between the unconditional and conditional reaction of term premia to movements in short-term yields, as well as the state dependence during periods of balance sheet tool usage, is a critical feature of our model. Our empirical findings show that the data is consistent with these key theoretical mechanisms.

4.4 Quantifying the Risk-Sharing Motive

The previous results show that some of the the model's predictions characteristic predictions are supported in the data. This reassures us that the model is well-suited for welfare analysis. But before considering counterfactual policy rules and the optimal design of conventional and unconventional policy, we next want to quantify the magnitude of the risk-sharing motive. In order to do so, we first utilize the following Lemma which allows us to characterize the risk-sharing motive in terms of observables rather than risk-weighted consumption dispersion.

Lemma 1 (Consumption dispersion and risk prices.)**.** *Assume $B = 0$. Consumption dispersion can be written as*

$$\check{\mathbf{c}}_t^\sigma = \frac{\rho}{1+\varsigma/\varphi}\mathbf{b}_t^\sigma \; - \; \varsigma^{-1} a \check{\mathbf{\Sigma}} \mathbf{P}_t, \qquad \mathbf{P}_t \equiv \mathrm{E}_t \int_0^\infty e^{-\rho s} \mathbf{\Lambda}_{t+s} \, \mathrm{d}s \,, \tag{50}$$

and the dispersion term of the loss (30) is therefore

$$\mathrm{Var}_i \, \check{c}_t(i) = \left(\frac{\rho}{1+\varsigma/\varphi}\right)^2 \mathbf{b}_t^{\sigma\top} \check{\mathbf{\Sigma}}^+ \mathbf{b}_t^\sigma - \frac{2\rho a}{(1+\varsigma/\varphi)\varsigma} \mathbf{b}_t^{\sigma\top} \mathbf{P}_t + \frac{a^2}{\varsigma^2} \mathbf{P}_t^\top \check{\mathbf{\Sigma}} \mathbf{P}_t. \tag{51}$$

In an affine equilibrium with $\mathbf{\Lambda}_t = \mathbf{L}\mathbf{s}_t$ with dynamics (44) we have $\mathbf{P}_t = \mathbf{L}(\rho\mathbf{I}+\mathbf{\Gamma})^{-1}\mathbf{s}_t$.

We obtain a simple closed-form approximation with the following simplifications: (i) assume $\mathrm{E}_t[a\mathbf{\Lambda}_{t+s}] = e^{-\kappa_p s} a\mathbf{\Lambda}_t$ for a scalar mean reversion rate κ_p; (ii) assume small risk-weighted wealth dispersion $\mathbf{b}_t^\sigma \approx 0$; (iii) assume a rank-one portfolio dis-

persion matrix $\check{\boldsymbol{\Sigma}} = s_0^2(\boldsymbol{\eta}^\sigma)^\top \boldsymbol{\eta}^\sigma$, *so that* $\tilde{\lambda}_t \equiv a\boldsymbol{\eta}^\sigma \boldsymbol{\Lambda}_t$. *Then*

$$\mathrm{E}\left[\mathrm{Var}_i\, \check{c}_t(i)\right] \approx \frac{s_0^2 \,\mathrm{Var}(\tilde{\lambda}_t)}{\varsigma^2 \left(\rho + \kappa_p\right)^2}. \tag{52}$$

With this approximation, the unconditional welfare terms unrelated to policy frictions are given by

$$\frac{1}{2}\mathrm{E}\left[\mathcal{L}_t\right] = \frac{\varsigma + \varphi}{2}\mathrm{Var}(x_t) + \frac{\vartheta}{2}\mathrm{Var}(\pi_t) + \frac{\varsigma}{2\varphi}\left(\varphi + \varsigma\right)\frac{s_0^2 \,\mathrm{Var}(\tilde{\lambda}_t)}{\varsigma^2 \left(\rho + \kappa_p\right)^2}.$$

Using a baseline calibration of $\rho = 0.02$ and $\varsigma = \varphi = 1$ and a relatively large rigidity term $\vartheta = 100$, we can calculate the welfare losses which arise from output gap and inflation volatility. Using US data from 1990-2019, measuring the output gap as deviations from potential and using either core PCE or CPI as the measure of inflation, the macroeconomic sources of welfare losses are $\approx [0.25, 0.5]$ (largely driven by the inflation loss terms). To compare this with the risk-sharing motive, we estimate the remaining terms from (52).

First, we measure average duration exposure and dispersion from the Survey of Consumer Finances. Assigning simple duration values to the main balance-sheet categories (deposits ≈ 0, bonds/bond funds ≈ 5, retirement 40/60 bond-equity, mortgages ≈ -5, equity $\in \{0, 4, 20\}$), the income-scaled exposure dispersion from the SCF microdata implies that $s_0 \approx [1.3, 1.8]$ (though depending on assumptions about equity duration, this value can be as large as $s_0 \approx 6.4$).

Next, using we measure term premia $\tilde{\lambda}_t$ either directly from Adrian et al. (2013), Kim and Wright (2005), or indirectly using excess returns from Gürkaynak et al. (2007). The annualized standard deviation of the duration-weighted market premia is $\approx [76bp, 101bp]$. Half-life estimates for term premia imply $\kappa_p \approx [0.2, 0.29]$. Thus, using the more conservative estimates for $s_0 \in [1.3, 1.8]$, we find that the risk-sharing motive is $\approx [0.18, 0.68]$. Therefore, even under conservative assumptions, the risk-sharing motive is at least comparable to the macroeconomic stability motives, and may be over twice as large. Note further that our analysis is based on duration-exposure-dispersion only, as is consistent with the model; extending the logic to other risky asset classes would imply even larger risk-sharing motives. We therefore view our estimates as a lower bound for the welfare costs of imperfect risk-sharing within the model, and so now turn to studying policies which seek to mitigate these costs.

5 Welfare Effects: Simple Policy Rules

Using the results of the previous section, we explore the welfare consequences and optimal design of simple policy rules. For now, we continue as in Section 3 to assume that the only aggregate source of uncertainty is natural rate shocks v_t, and that prices are fully rigid.

We first study simple policy rules which (in equilibrium) are only a function of the "natural" state variables; under the simplifying assumptions considered thus far, these are natural rate shocks v_t (equivalently, technology shocks z_t) and the risk-weighted household wealth distribution b_t^σ. Thus, we study policy rules which implement

$$i_t = \chi_{i,v} v_t + \chi_{i,b} b_t^\sigma, \tag{53}$$

$$qe_t(\tau) = \chi_{q,v}(\tau) v_t + \chi_{q,b}(\tau) b_t^\sigma \implies qe_t^\sigma = \chi_{q,v}^\sigma v_t + \chi_{q,b}^\sigma b_t^\sigma, \tag{54}$$

for some choice of policy parameters $\chi_{i,v}, \chi_{i,b}$ and $\{\chi_{q,v}(\tau), \chi_{q,b}(\tau)\}_{\tau=0}^T$. We assume that such a policy is implementable; determinacy conditions can be satisfied through a Taylor rule like those considered in Section 3, which generate the unique "minimum state variable" equilibrium as in McCallum (1983). Note that in this context with a single priced risk factor, for the same reasons as discussed in Section 2 the impact of balance sheet rules across the term structure can be fully summarized by the risk-weighted balance sheet object qe_t^σ, up to considerations of policy frictions $\psi(\tau)$, which the following Lemma makes precise.

Lemma 2 (Optimal QE maturity profile)**.** *Equilibrium dynamics and welfare depend on central bank holdings* $\{qe_t(\tau)\}_{\tau=0}^T$ *only through the risk-weighted vector* $\mathbf{qe}_t^\sigma$ *and the holding cost* $\int_0^T \psi(\tau) qe_t(\tau)^2 \, \mathrm{d}\tau$*. Given* $\mathbf{qe}_t^\sigma$*, the optimal portfolio therefore solves a simple minimum cost problem when* $\psi(\tau) > 0$*:*

$$qe_t(\tau) = \frac{\boldsymbol{\sigma}(\tau) \, \boldsymbol{\Omega}_\psi^{-1} \mathbf{qe}_t^\sigma}{\psi(\tau)}, \quad \boldsymbol{\Omega}_\psi \equiv \int_0^T \frac{\boldsymbol{\sigma}(\tau)^\top \boldsymbol{\sigma}(\tau)}{\psi(\tau)} \, \mathrm{d}\tau \,, \tag{55}$$

and the flow costs of the policy are $\int_0^T \psi(\tau) \, qe_t(\tau)^2 \, \mathrm{d}\tau = \mathbf{qe}_t^{\sigma\top} \boldsymbol{\Omega}_\psi^{-1} \mathbf{qe}_t^\sigma$*.*

Lemma 2 formalizes the extent of "localization" in our model which the policymaker may exploit. Despite a continuum of instruments, the policymaker can only effectively control the market price of risk, which is inherently low-dimensional due to arbitrageur-implied tight restrictions between risk prices and term premia across

maturities. In particular, in our stylized model considered here, balance sheet rules work through only a single risk factor.

The policymaker chooses policy parameters in order to minimize unconditional social loss

$$\mathcal{W} \equiv \frac{1}{2}\mathbb{E}\int_0^\infty e^{-\rho t}\mathcal{L}_t \, \mathrm{d}t \,,$$

where per-period social welfare loss $\mathcal{L}_t$ is derived in (29)-(31). Thus, we consider a policymaker who is able to commit to simple policy rules, which are functions of the natural state variables of the economy. In the next section, we explore optimal policy for a policymaker who conducts (history-dependent) policy under full commitment.

5.1 Optimal Policy: Short Rate Only

First, consider the benchmark case of a risk neutral arbitrageurs: $a = 0$. Then the expectations hypothesis holds, so regardless of the asset positions of households and firms (or central bank balance sheet policies), borrowing rates are equalized: $\mu_t(\tau) = i_t$. Moreover, as long as initial household wealth is equalized, the model collapses to the standard RANK case. In particular, consumption and wealth dispersion disappear: $b_t^\sigma = 0, \quad \mathrm{Var}_i \, \check{c}_t(i) = 0 \quad \forall t$. Because inflation is zero due to rigid prices (or more generally, when divine coincidence holds), the conventional policy rule with $\chi_{i,v} = 1$ which implements $i_t = v_t$ achieves first-best: $x_t = 0$ for all periods t. We further recover the standard QE neutrality result: balance sheet policies do not affect bond prices (and therefore have no aggregate effects).

Under risk-neutrality, so long as the central bank faces no costs to setting the policy rate, first-best is achievable with only conventional policy. The next Proposition shows that these results fail whenever arbitrage is imperfect ($a > 0$); this is true even if absent the consumption risk-sharing motive. We derive optimal short-rate policy when the central bank does not have access to balance sheet tools (formally, when balance sheet costs $\psi(\tau) \to \infty$).

Proposition 6 (Optimal short rate policy rule)**.** *Assume risk aversion $a > 0$, $\eta^\sigma > 0$, and $\beta^\sigma > 0$. If bond risk dispersion across households $\check{\Sigma} = 0$, then some $\chi^n_{i,v} \lesseqgtr 1$ and $\chi_{i,b} = 0$ in (53) guarantees $x_t = 0$ each period. The sign of $\chi^n_{i,v} - 1$ is determined by the endogenous reaction of firm borrowing to natural rate shocks.*

With $\check{\Sigma} > 0$, the optimal short rate policy rule is given by $i_t = \chi^*_{i,v} v_t + \chi_{i,b} b^\sigma_t$ where generically $\chi_{b,i} \neq 0$ and $\chi^*_{i,v} < \chi^n_{i,v}$. The optimal short rate policy implies:

1. *Bond carry trade returns $\mu_t(\tau) - i_t$ move in the same direction as i_t iff firm borrowing declines in response to natural rate shocks.*
2. *Output gaps x_t are not identically zero.*
3. *Consumption dispersion is non-zero:* $\mathrm{Var}_i\, \check{c}_t(i) \neq 0$.

We can understand this result using the intuition derived in the previous section. Consider a fall in the natural rate inducing a cut in the policy rate. If firm borrowing declines, then arbitrageurs reduce their position in the bond carry trade, which requires a smaller bond carry trade return. On the other hand, if instead firm borrowing increase, through market clearing, bond arbitrageurs increase their position in the bond carry trade, which requires a larger bond carry trade return.

Because of this, risk premia vary over time. Thus, a simple short rate policy is unable to equalize all borrowing rates. If the risk-weighted dynamics of household wealth b^σ_t are negligible, then the central bank can choose $i_t = \chi^n_{i,v} v_t$, which implies that $\tilde{\mu}_t = v_t$ (and therefore closes output gaps). However, fluctuations in the natural rate induce volatile fluctuations in the policy rate, and the volatility of short rate changes is increasing in the responsiveness of conventional policy to natural rates. More volatile short rates implies greater variation in term premia. Individual Euler equations differ, which implies that consumption choices across households $c_t(i) \neq c_t(i')$. Therefore, consumption dispersion $\mathrm{Var}_i\, \check{c}_t(i) \neq 0$. Thus, from a risk-sharing perspective, choosing a policy rule which is less reactive to natural rate fluctuations is welfare-improving.

Of course, if the short rate becomes less responsive to natural rates, output gap volatility increases. For very small choices of $\chi_{i,v}$, consumption dispersion becomes negligible. However, output gap volatility increases substantially. The optimal choice of $\chi^*_{i,v}$ balances these forces. At the optimum, shocks to the natural rate do not fully pass through to the effective borrowing rate: $\tilde{\mu}_t \neq v_t$. Thus, aggregate borrowing demand changes, and hence the output gap $x_t \neq 0$.

Finally, when the risk-weighted dynamics of household wealth b_t^σ are non-negligible, it is not possible for the central bank to choose any $i_t = \chi_{i,v}^n v_t$, which implies that $\tilde{\mu}_t = v_t$. Because fluctuations in b_t^σ affect term premia, then even conditional on the level of the natural rate, term premia will fluctuate through the endogenous movements in household wealth. Thus, the optimal policy in this case must place some weight on risk-weighted household wealth b_t^σ.

5.2 Optimal Short Rate and Balance Sheet Policy

The failure of conventional policy to achieve first-best is driven by three frictions. First, because of imperfect pass-through of the policy rate to household borrowing rates, natural rate shocks are not fully accommodated, implying excessive volatility of output. Second, borrowing rates across households differ, implying suboptimal consumption dispersion. Third, the dispersion in consumption decisions induces additional equilibrium movements in relative bond positions of households, leading to further fluctuations in term premia. Simple conventional policy rules cannot overcome all frictions simultaneously. However, when the central bank has access to frictionless balance sheet policies, we obtain the following result.

Proposition 7 (Optimal policy separation principle)**.** *Assume risk aversion $a > 0$, $\eta^\sigma > 0$, and $\beta^\sigma > 0$. Further, assume that household wealth is initially equalized. Suppose the central bank implements short rate and balance sheet policy according to*

$$i_t = v_t, \quad qe_t^\sigma = -\frac{\beta^\sigma}{\varsigma \kappa_z} v_t.$$

If short rate policy is frictionless ($\psi^i = 0$) and the central bank does not face holding costs ($\psi(\tau) = 0$), then first-best is achieved:

1. *Macroeconomic stabilization: $x_t = 0 \; \forall t$.*
2. *Term premia stabilization: $\mu_t(\tau) = \tilde{\mu}_t \; \forall \tau$.*
3. *Consumption equalization: $c_t(i) = c_t(i') \; \forall i, i'$.*

The results follow naturally from our findings regarding ad-hoc policy. The balance sheet policy in Prop. 7 stabilizes shocks to bond markets by offsetting all firm borrowing shocks. The policy implies that risk premia are zero and thus equalizes

borrowing rates across households: $\mu_t(\tau) = \tilde{\mu}_t$. When this is the case, the model collapses to a standard RANK model. The elimination of risk premia implies that policy shocks are transmitted one-to-one to borrowing rates, so $\tilde{\mu}_t = i_t$. Thus, the policy $i_t = v_t$ implies output gaps $x_t = 0$ which is optimal as inflation is zero by assumption (but will also be optimal in general when divine coincidence holds).

Thus, we derive an optimal *separation principle* for optimal policy: optimal balance sheet policy *stabilizes long-term rates* while optimal short rate policy *stabilizes macroeconomic aggregates*.

5.3 Optimal Policy: Balance Sheet Only

We now derive the optimal use of balance sheet tools when the central bank faces constraints on the short rate. To capture the essence of short rate constraints in a simple way without introducing an explicit ZLB, we assume that $\psi^i \to \infty$ and assume a simple process for the "target rate" $\bar{i}_t$ in (20). The next Proposition shows that balance sheet tools alone are not enough to achieve first-best.

Proposition 8 (Optimal balance sheet rule)**.** *Assume risk aversion $a > 0$, $\eta^\sigma > 0$, and $\beta^\sigma > 0$. Further, assume that household wealth is initially equalized. Suppose that $\psi^i \to \infty$ and that $\bar{i}_t = \tilde{\chi}_i v_t$ where $0 < \tilde{\chi}_{i,v} < 1$. Consider the balance sheet rule which implements*

$$qe_t^\sigma = \beta^\sigma \left((1 + \varsigma + \varphi) x_t - \frac{1}{\varsigma \kappa_z} v_t \right).$$

This policy equalizes borrowing rates ($\mu_t(\tau) = \tilde{\mu}_t \ \forall \tau$), and consumption and wealth dispersion are zero: $\mathrm{Var}_i \check{c}_t(i) = 0$ and $b_t^\sigma = 0$. However, output gaps x_t are no longer identically zero.

There exists some balance sheet policy parameters $\chi_{q,v}^\sigma \neq 0, \chi_{q,b}^\sigma \neq 0$ such that $\tilde{\mu}_t = v_t$ which satisfies macroeconomic stabilization ($x_t = 0 \ \forall t$). However, long-term bond return stabilization fails ($\exists \tau : \mu_t(\tau) \neq \tilde{\mu}_t$), and therefore consumption and wealth dispersion are non-zero. The optimal balance sheet rule implies:

1. *Output gaps x_t are not identically zero. Relative to no balance sheet policies, output gap volatility is lower.*

2. *Consumption dispersion is non-zero: $\mathrm{Var}_i \check{c}_t(i) \neq 0$. Relative to no balance*

sheet policies, consumption dispersion is lower than under full macroeconomic stabilization via balance sheet tools.

Prop. 8 shows that when the short rate is constrained, the optimal balance sheet policy must sacrifice term premia stabilization in order to (partially) stabilize macroeconomic volatility. While balance sheet tools can continue to equalize borrowing rates, suboptimal short rate policy implies that the effective household borrowing rate $\tilde{\mu}_t \neq v_t$. Thus, this policy does not achieve macroeconomic stabilization: $x_t \neq 0$. On the other hand, balance sheet policies alone can close the output gap, but this is also suboptimal because borrowing rates are no longer equalized.

It may be surprising that balance sheet tools are insufficient to simultaneously achieve macroeconomic stabilization and bond market stabilization. Although we have eliminated the short rate as an active policy tool, we have given the policymaker a continuum of balance sheet tools $\{qe_t(\tau)\}_{\tau=0}^{T}$. The usual textbook intuition seems to suggest that the policymaker has enough tools to deal with the two objectives.

The reason for the failure of textbook intuition is that balance sheet policy works by affecting term premia through changes in the market price of risk. Although arbitrage is imperfect in this model, arbitrageurs still enforce tight restrictions between between the market price of risk and term premia across maturities. Hence, while in principle the central bank has a continuum of policy tools, in practice it can only manipulate λ_t (which we have already imposed by working with the risk-weighted balance sheet object qe_t^{σ}).

This is related to the localization results in Vayanos and Vila (2021) and Ray et al. (2024). In the one-factor model considered here, the effects of QE are fully global. However, even with more complicated risk structure, localization is not strong enough to allow balance sheet policy rules alone to achieve first-best.

5.4 Optimal Policy without Divine Coincidence

Without policy frictions, the optimal separation principle of Prop. 7 shows that when divine coincidence holds, the central banker can achieve first-best. We now study policy when divine coincidence fails. We maintain the stylized assumptions of the model so far, except we relax the assumption of fully rigid prices, and we allow for nonzero markup shocks $u_t = u_z z_t$ (for simplicity, we maintain the one-factor assumption, so that markup shocks load on the single source of aggregate risk). Under

this relaxed model, we regain our augmented Phillips curve and have

$$\begin{aligned}
\mathrm{E}_t\,\mathrm{d}x_t &= \varsigma^{-1}\left[i_t + a\eta^\sigma\Lambda_t - \pi_t - v_t\right]\mathrm{d}t\,,\\
\mathrm{E}_t\,\mathrm{d}\pi_t &= \left[\rho\pi_t - \kappa\left(x_t + \alpha_{\hat{\mu}}\hat{\mu}_t\right) - u_t\right]\mathrm{d}t\,,\\
\hat{\mu}_t &= \beta i_t + a\beta^\sigma\Lambda_t.
\end{aligned}$$

All other variables and endogenous coefficients remain the same. Define the firm-household effective-rate wedge

$$\Delta^\sigma \equiv \beta^\sigma - \beta\eta^\sigma, \tag{56}$$

which measures the relevant distance between the firm effective borrowing rate and the aggregate household effective borrowing rate. When $\Delta^\sigma \neq 0$, in principle it is possible for the central bank to utilize short rate and balance sheet tools to separately target these borrowing rates. We explore the feasibility of simultaneously stabilizing the output gap and inflation and the welfare costs in the next Proposition.

Proposition 9 (Exact macro stabilization.)**.** *Suppose that* $a > 0, \beta > 0, \beta^\sigma > 0, \eta^\sigma > 0$.

1. *Whenever* $\Delta^\sigma \neq 0$*, exact macro stabilization* $x_t \equiv \pi_t \equiv 0$ *is achieved via*

$$\Lambda_t = -\frac{\nu_t}{a\Delta^\sigma},\quad i_t = v_t + \frac{\eta^\sigma}{\Delta^\sigma}\nu_t,\quad qe_t^\sigma = -\frac{\beta^\sigma}{\varsigma\kappa_z}v_t - b_t^\sigma - \Lambda_t, \tag{57}$$

   where $\nu_t = \beta v_t + \frac{\vartheta}{\epsilon}u_t$ *(so long as* $\nu_t \not\equiv 0\ \forall t$*).*

2. *If* $\check{\Sigma} > 0$*, then as* $\Delta^\sigma \to 0$*, the social welfare costs of* (57) *become unbounded. Moreover, the policy tools themselves diverge.*

Typically, the presence of a working capital channel in the Phillips curve endogenously generates the failure of divine coincidence. However, in our model, because firm and household borrowing costs differ, policymakers may be tempted to utilize a combination of short rate and balance sheet tools to separately target these effective rates. With two policy tools this is possible: the policymaker manipulates both the short rate and term premia to set both the household and firm effective rates to any desired target. This not only allows the central bank to offset the endogenous "cost channel" of conventional monetary policy, but in fact allows for a full offset of any

shocks to the Phillips curve. But such a policy comes at a price: because the policy necessarily involves manipulating risk prices, it worsens risk sharing. In the limit as household and firm effective rates become more similar, the required policy leads to unboundedly large fluctuations in term premia, and therefore the welfare costs of such a policy themselves become unbounded.

5.5 Optimal Policy Calibration

We now study quantitatively the implications for simple policy rules in a calibrated model. We extend the model to include three factors: discount factor shocks, markup shocks, and a noise demand shock. We use standard externally calibrated parameters where possible, using either textbook New Keynesian parameterizations, parameterizations from Greenwood and Vayanos (2014), or reading directly from the the welfare calculations in Section 4.4. We have five parameters which are internally calibrated: the volatility of each shock, the correlation between the discount factor shock and the noise-demand shock, and arbitrageur risk aversion. We target the following moments: the volatility of the output gap, inflation, the correlation of the output gap and inflation, term premia volatility, and the correlation of short rate changes and term premia changes.

We then study the macroeconomic and welfare consequences of counterfactual policy designs.

Table 1: Counterfactual Policy Rules

Rule	$\mathcal{L} \times 10^3$	SD(x)	SD(π)	premium	$\hat{a}_{\max}/\hat{a}_{\text{cal}}$
historical Taylor $(1.5, 0.5)$	3.60	1.85%	0.38%	59bp	1.15
gap-stabilized Taylor ($\phi_x \to \infty$)	3.39	0.00%	0.31%	79bp	1.18
π-stabilized Taylor ($\phi_\pi = 5$)	5.07	1.76%	0.33%	96bp	1.00
Λ_0-QE + i-rule $(1.0, 0.0)$	0.93	0.12%	0.30%	0	∞
Λ_0-QE + i-rule $(1.10, 1.5)$	**0.91**	0.44%	0.30%	0	∞
Λ_0-QE + gap-stabilized Taylor	0.95	0.00%	0.31%	0	∞

Table 1 reports our results. Each row corresponds to a different policy rule. The top panel considers short-rate-only rules, while the bottom panel additional considers QE. The first column reports the social welfare loss of the given rule; the next columns report output gap, inflation, and term premia volatility; the final column reports the

maximum arbitrageur risk aversion for which the equilibrium exists (relative to our estimates).

The first row corresponds to our estimated model using a historical Taylor rule. We next consider policies which seek to fully stabilize the output gap or inflation. The second column shows that closing the output gap only modestly improves welfare; inflation volatility also falls, but term premia volatility goes up, so that overall the policy only minimally improves welfare.

The next row shows that inflation stabilization through a larger Taylor coefficient proves unsuccessful: because of the working capital channel, a large response to inflation is necessary to further decrease inflation volatility; but the increased volatility of the short rate pushes up term premia volatility, which worsens the working capital channel. In our calibration, a Taylor coefficient of 5 implies that the risk premia volatility is maximal; further increases in responsiveness imply a breakdown of the affine equilibrium as arbitrageurs no longer intermediate bond markets.

The second panel considers simple quasi-optimal rules using both the short rate and QE. In each case, we consider the QE rule which fully offsets term premia, and therefore fully removes the costs from imperfect risk-sharing. We see that the welfare improvements are massive. Different short rate rules change the welfare calculations somewhat, but these are largely through different output gap volatilities, which as is standard have only small welfare consequences. The remaining welfare costs come from inflation volatility, which in general will remain due to the endogenous failure of divine coincidence from the working capital.

Table 2: Counterfactual Policy Implementation

Rule	$\boldsymbol{\iota}$	v-response	u-response
historical Taylor	$(0.218,\ 1.519,\ -0.003,\ -0.118)$	0.87	1.52
gap-stab Taylor	$(0.261,\ 1.166,\ -0.005,\ -0.081)$	1.05	1.17
π-stab Taylor ($\phi_\pi = 5$)	$(0.237,\ 3.823,\ -0.005,\ -0.118)$	0.95	3.82
best $\{v, u, s\}$	$(0.240,\ 0.329,\ -0.004,\ -0.121)$	0.96	0.33
Λ_0-QE + $(1.0, 0.0)$	$(0.262,\ 1.518,\ 0,\ 0)$	1.05	1.52
Λ_0-QE + $(1.10, 1.5)$	$(0.267,\ 2.426,\ 0,\ 0)$	1.07	2.43

Next, Table 2 looks at the implied equilibrium loading on the state variables: discount factor shocks, markup shocks, noise demand, and wealth dispersion. The final two columns report the loading on the transformed natural rate v and the markup

shock. For each policy, we see that the implied on-equilibrium response to natural rates is not that different from the RANK benchmark of 1. What differs substantially is the implied response to markup shocks. The larger responses to markup shocks imply larger risk-sharing losses; as we can see in the "best $\{u, v, s\}$" row, the constrained optimal short rate policy significantly under-reacts to markup shocks relative to the historical benchmark. The under-reaction which we discussed in the stylized analytical model carries over here. However, once balance sheet tools are utilized effectively, and the short rate is able to respond aggressively once more without a deterioration in risk-sharing.

6 Welfare Effects: Full Commitment

We now study full commitment: the policymaker chooses short rate and balance sheet policies as functions of the entire history of shocks, minimizing the conditional social loss

$$\mathcal{W}_t = \mathbb{E}_t \int_t^\infty \frac{1}{2} e^{-\rho s} \mathcal{L}_s \, \mathrm{d}s \,. \tag{58}$$

Our optimal separate result in Prop. 7 show that simple policy rules can achieve first-best when the policymaker faces no short rate or balance sheet frictions and divine coincidence holds. But outside of these cases, full commitment policies can improve on simple policy rules.

We consider the problem of a planner with full-commitment minimizing 58; the problem is unrestricted, except that we restrict ourselves to the class of models which feature time-invariant bond loadings $\boldsymbol{\sigma}(\tau)$. We therefore study models with an extended class of state variables, augmenting with a set of *policy memory states* $\mathbf{m}_t$:

$$i_t = \boldsymbol{\phi}_i^\top \mathbf{s}_t + \boldsymbol{\phi}_{im}^\top \mathbf{m}_t + \epsilon_{i,t}, \quad \mathbf{qe}_t^\sigma = \boldsymbol{\Phi}_q \mathbf{s}_t + \boldsymbol{\Phi}_{qm} \mathbf{m}_t + \boldsymbol{\epsilon}_{q,t}, \tag{59}$$

where

$$\mathrm{d}\mathbf{m}_t = -\left[\boldsymbol{\Gamma}_m \mathbf{m}_t + \boldsymbol{\Gamma}_{ms} \mathbf{s}_t\right] \mathrm{d}t + \mathbf{K}_m \, \mathrm{d}\mathbf{B}_t \,, \tag{60}$$

with every object (including the dimension of $\mathbf{m}_t$) is a choice of the planner, along with the deterministic paths $(\epsilon_{i,t}, \boldsymbol{\epsilon}_{q,t})$. Bond prices are then affine in the extended

state $\mathbf{s}_t^+ \equiv (\mathbf{s}_t^\top, \mathbf{m}_t^\top)^\top$ plus a deterministic intercept. Crucially, all elements of the extended state $\mathbf{s}_t^+$ are priced risk factors; the deterministic paths move expected returns without affecting loadings or risk pricing.

Two results characterize the optimal plan. The following Lemma temporarily takes the endogenous risk-weighted composites $(\boldsymbol{\eta}^\sigma, \boldsymbol{\beta}^\sigma, \check{\boldsymbol{\Sigma}}, \boldsymbol{\Omega}_\psi)$ fixed. The problem is then a fully standard linear-quadratic program, and hence its solution takes the familiar form of "promise-keeping" costates.

Lemma 3 (Commitment with exogenous volatility)**.** *Given the bond volatility loadings $\boldsymbol{\sigma}(\tau)$, the full-commitment optimum is characterized by the planner's first-order conditions in which the costates on forward-looking equations act as promise states:*

$$\begin{aligned}
\mathrm{d}\theta_{x,t} &= [\rho\theta_{x,t} - (\varsigma+\varphi)x_t + \kappa\theta_{\pi,t} - (\varsigma+\varphi)\theta_{B,t} - (1+\varsigma+\varphi)\,\boldsymbol{\beta}^\sigma\boldsymbol{\theta}_{\Lambda,t}]\,\mathrm{d}t\,,\\
\mathrm{d}\theta_{\pi,t} &= \left[-\vartheta\pi_t + \varsigma^{-1}\theta_{x,t} + B\theta_{B,t}\right]\mathrm{d}t\,,\quad \mathrm{d}\boldsymbol{\theta}_{c,t} = \left[\rho\boldsymbol{\theta}_{c,t} - \frac{\varsigma(\varsigma+\varphi)}{\varphi}\check{\boldsymbol{\Sigma}}^+\check{\mathbf{c}}_t^\sigma + (1+\varsigma/\varphi)\,\boldsymbol{\theta}_{b,t}\right]\mathrm{d}t\,,
\end{aligned} \tag{61}$$

started at $\theta_{x,0} = \theta_{\pi,0} = 0$, $\boldsymbol{\theta}_{c,0} = \mathbf{0}$, where $\boldsymbol{\theta}_{\Lambda,t}$ is the shadow value of the risk price (a linear combination of the costates) and $(\boldsymbol{\theta}_{b,t}, \theta_{B,t})$ are the costates on the wealth states, pinned by boundedness. The optimal instruments are linear in the natural states and the promise states; in particular, policy is inertial, responding to innovations through dynamics of the promise costates.

The second result removes the restriction to fixed composites, and takes into account that the fully optimal planner internalizes that the bond volatilities $\boldsymbol{\sigma}(\tau)$ (and hence risk transmission, portfolio dispersion, and balance sheet costs) themselves depend on policy.

Theorem 3 (Optimal policy under full commitment)**.** *Denote the multipliers $\boldsymbol{\nu}(\tau) \in \mathbb{R}^{N_z}$ associated with endogenous volatility loading $\boldsymbol{\sigma}(\tau)$, and define the* shadow factors

$$\upsilon_t^i \equiv \int_0^\infty \boldsymbol{\nu}_i(u)^\top\,\mathrm{d}\mathbf{B}_{t-u}\,,\quad \boldsymbol{\upsilon}_t^\Lambda \equiv \int_0^\infty \boldsymbol{\nu}_\Lambda(u)^\top\,\mathrm{d}\mathbf{B}_{t-u}\,, \tag{62}$$

$$\boldsymbol{\nu}_i(u) \equiv \int_u^T \boldsymbol{\nu}(\tau)\,\mathrm{d}\tau\,,\quad \boldsymbol{\nu}_\Lambda(u) \equiv a\int_u^T \boldsymbol{\nu}(\tau)\,\boldsymbol{\sigma}(\tau-u)\,\mathrm{d}\tau\,. \tag{63}$$

The optimum is characterized by the system of Lemma 3 with $\boldsymbol{\theta}_{\Lambda,t}$ replaced by $\boldsymbol{\theta}_{\Lambda,t}+\boldsymbol{\upsilon}_t^\Lambda$ and the short-rate condition amended by υ_t^i.

Let $c \in \mathbb{R}^K$ collect the distinct entries of all endogenous risk-weighted composites. The multiplier function is spanned by K basis functions,

$$\boldsymbol{\nu}(\tau) = \sum_{k=1}^{K} \vartheta_k \boldsymbol{\nu}^{(k)}(\tau), \quad \vartheta_k \equiv \frac{\partial J}{\partial c_k}, \tag{64}$$

where $J(c)$ denotes the optimal value of the planner's problem holding the composites fixed at c. Hence, the optimum is characterized by the finite-dimensional fixed point problem in $2K$ coefficients $(c, \boldsymbol{\vartheta})$. The optimal instruments take the form

$$i_t = \epsilon_{i,t} + \boldsymbol{\phi}_i^\top \mathbf{s}_t + \boldsymbol{\phi}_{i\theta}^\top \boldsymbol{\theta}_t + \int_0^\infty \mathbf{f}_i(u)^\top \, \mathrm{d}\mathbf{B}_{t-u} \,, \tag{65}$$

$$\mathbf{qe}_t^\sigma = \boldsymbol{\epsilon}_{q,t} + \boldsymbol{\Phi}_q \mathbf{s}_t + \boldsymbol{\Phi}_{q\theta} \boldsymbol{\theta}_t + \int_0^\infty \mathbf{F}_q(u) \, \mathrm{d}\mathbf{B}_{t-u} \,, \tag{66}$$

with convolution kernels linear in $\boldsymbol{\vartheta}$. The construction and algorithm are in Appendix B.4.

Commitment improves welfare through a novel mechanism in our model. As in standard New Keynesian commitment problems, promises induce inertial, "price-level-targeting"-style policy responses. Here, inertia has an additional payoff: less reactive policy implies *ex-ante* smaller expected variation in policy tools, which in equilibrium reduces endogenous bond volatility and hence the risk compensation arbitrageurs require. Thus, inertial policy additionally improves risk-sharing directly. The shadow factors in Theorem 3 capture a distinct *term-structure-shaping* motive: the planner values impulse responses of the short rate and risk prices because they determine the endogenous loadings, not because direct impacts on the loss function. Moreover, the problem is fully solved and computable, as described in the algorithm in Appendix B.4. Quantitative analysis is left to future numerical work.

7 Measuring Policy Objectives: Predictability

We now consider simple observable tests related to the optimality of balance sheet policies. We consider the bond return predictability regressions of Fama and Bliss

(1987) (FB):

$$\frac{1}{\Delta\tau}\log\left(\frac{P_{t+\Delta\tau}^{(\tau-\Delta\tau)}}{P_t^{(\tau)}}\right) - y_t^{(\Delta\tau)} = a_{FB}^{(\tau)} + b_{FB}^{(\tau)}\left(f_t^{(\tau-\Delta\tau,\tau)} - y_t^{(\Delta\tau)}\right) + \varepsilon_{t+\Delta\tau}.$$

These regression coefficients measure how the slope of the term structure predicts excess returns. Figure 7 shows the well-known result that (in the full sample), Fama-Bliss coefficients are positive, and an increasing function of maturity.

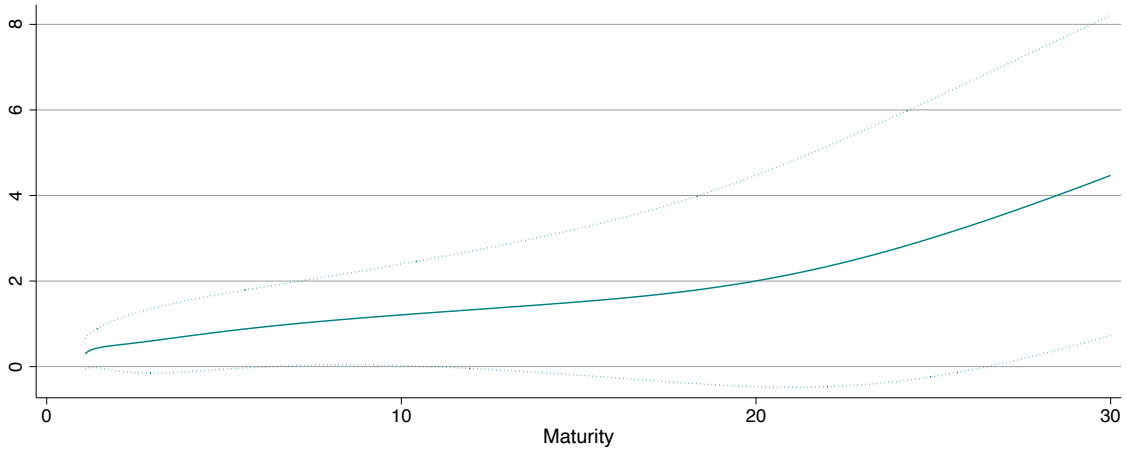


Figure 7: Fama-Bliss Coefficients: Treasuries, Full Sample

In our model, when the central bank does not use the balance sheet and only relies on conventional short rate policy, then

$$\bar{b}_{FB}^{(\tau)} > 0.$$

However, if balance sheet policy is successfully pursuing term premia stabilization, then the implied FB coefficient shrinks towards zero:

$$\bar{b}_{FB}^{(\tau)} > \hat{b}_{FB}^{(\tau)} \to 0.$$

Instead, if balance sheet policy is pursuing macroeconomic stabilization, we have that the estimated FB coefficient is larger than under conventional policy only:

$$\hat{b}_{FB}^{(\tau)} > \bar{b}_{FB}^{(\tau)}.$$

We examine these stylized predictions by studying how FB regression coefficients vary as a function of different monetary policy regimes.

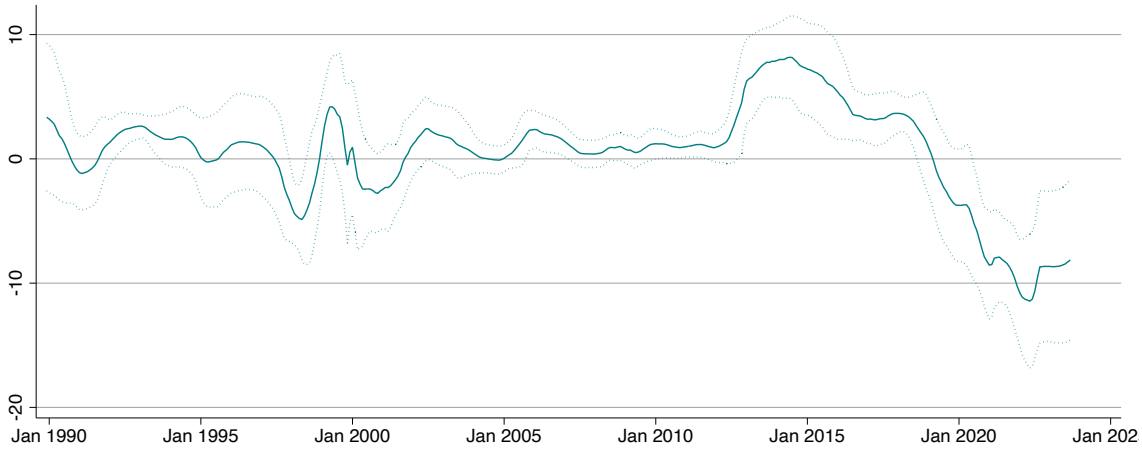


Figure 8: Fama-Bliss Coefficients: 10-year Treasuries, Rolling Sample

Figure 8 estimates rolling Fama-Bliss coefficients (fixing $\tau = 10$ as a baseline maturity). We see that Fama-Bliss coefficients increased substantially during the initial QE regime, but have fallen and even become negative in recent years. This is consistent with the predictions of the model if QE was initially undertaken for purely macroeconomic stabilization purposes. However, these findings imply that in the most recent years since conventional policy has become unconstrained, the mix of conventional and unconventional tools is not optimal.

8 Concluding Remarks

This paper develops a tractable general equilibrium model with market segmentation, nominal frictions, and household heterogeneity. We first derive results for how monetary policy (conventional and unconventional) affects aggregate dynamics. We next show that welfare losses arise from excessive volatility in inflation, output, and excess consumption dispersion across households. The frictions associated with consumption dispersion arise due to market segmentation and excessive volatility of term premia. Optimal short rate and balance sheet policies are characterized by a sharp separation result: conventional policy targets macroeconomic stability, while unconventional policy stabilizes bond returns. Optimal policy removes excess volatility of

term premia and hence improves risk-sharing across households, while reducing excess macroeconomic volatility. Policy constraints on either the short rate or balance sheets imply tradeoffs between these two policy objectives.

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A Proofs

All proofs use the perturbation of Section 2.2: variables denote normalized first-order deviations from the deterministic steady state (DSS); products of first-order objects are second order and dropped; bond return loadings $\boldsymbol{\sigma}(\tau)$ are constant at first order; welfare is evaluated to second order along first-order-accurate dynamics. We omit standard transversality arguments when selecting unique bounded solutions of the model.

In some results for the one-factor model, we make use of additional perturbation method: "at leading order in a" means the result is based on an expansion in $a = 0$ (which is separate from the ξ-perturbation). We utilize this when determining signs: by continuity, any strictly signed leading-order term holds in some neighrborhood of $a = 0$.

Proof of Proposition 1. The optimal subsidy is

$$1 + \tau^y = \frac{\epsilon}{\epsilon - 1}(1 + \beta\rho), \tag{A1}$$

correcting both the steady-state markup and the working-capital wedge ($\hat{\mu} = \beta\rho$, since deterministic returns force $\mu(\tau) = i = \rho$ for all τ). *RA equivalence:* with $a = 0$, (19) forces $\mu_t(\tau) = i_t$ for all τ, hence $\tilde{\mu}_t(i) = i_t$ for every i: households face identical budget sets and, with equalized initial wealth, make identical choices. *First best:* with $\vartheta = 0$, firm optimality gives MC $= (\epsilon - 1)(1 + \tau^y)/\epsilon = 1 + \beta\rho$; since MC $= (1 + \hat{\mu})W/Z$, the real wage is efficient ($W = Z$), and (8) plus market clearing imply the allocation solves the equal-weight planner problem $\max u(C) - v(N)$ s.t. $C = ZN$. *Steady state:* under $i = \rho$ the stationary Euler equation gives $\pi = 0$; with $S(\tau) = B\eta(\tau) - \beta(\tau)$, clearing (23) gives $X(\tau) = 0$ pointwise, and equal wealth plus equal transfers equalize allocations. None of these conditions involve ϑ or a, so the sticky-price, imperfect-arbitrage economy shares this efficient steady state. □

Proof of Theorem 1. (1) Every household-specific term in (35)-(36) is proportional to $\boldsymbol{\eta}_i^\sigma - \boldsymbol{\eta}^\sigma$, so the span $\{(\boldsymbol{\eta}_i^\sigma - \boldsymbol{\eta}^\sigma)\mathbf{v} : \mathbf{v} \in \mathbb{R}^{N_z}\}$ is invariant under the dynamics and spanning propagates from $t = 0$; matching coefficients delivers (38). (2) Multiply (35) and (36) by $(\boldsymbol{\eta}_i^\sigma)^\top$ and integrate over i, using $\int_0^1 (\boldsymbol{\eta}_i^\sigma)^\top (\boldsymbol{\eta}_i^\sigma - \boldsymbol{\eta}^\sigma)\,\mathrm{d}i = \check{\boldsymbol{\Sigma}}$; the representations then follow from (1). (3) From (1)-(2); the second form in (39) holds because $\check{\mathbf{c}}_t^\sigma = \check{\boldsymbol{\Sigma}}\boldsymbol{\gamma}_t^c$ lies in the range of $\check{\boldsymbol{\Sigma}}$. (4) Bond market clearing (23) in first-order deviations is $x_t(\tau) = \beta(\tau)(w_t + n_t) + s_t(\tau) - qe_t(\tau) - \eta(\tau)b_t - \int_0^1 \eta_i(\tau)\check{b}_t(i)\,\mathrm{d}i$, and intratemporal optimality with production and goods clearing give the wage bill $w_t + n_t = (1 + \varsigma + \varphi)x_t + \frac{1+\varphi}{\varsigma+\varphi} z_t$. Integrating against $\boldsymbol{\sigma}(\tau)^\top$ gives (40). □

Proof of Proposition 2. Ito's lemma applied to (42) gives $\boldsymbol{\sigma}(\tau) = -\mathbf{A}(\tau)^\top \boldsymbol{\sigma}_s$ and drift $\mu_t(\tau) = [\mathbf{A}'(\tau) + \boldsymbol{\Gamma}^\top \mathbf{A}(\tau)]^\top \mathbf{s}_t + \text{const}$. Substituting into $\mu_t(\tau) - i_t = a\boldsymbol{\sigma}(\tau)\boldsymbol{\Lambda}_t$ with

$i_t = \boldsymbol{\iota}^\top \mathbf{s}_t$, $\boldsymbol{\Lambda}_t = \mathbf{L}\mathbf{s}_t$ and matching coefficients on $\mathbf{s}_t$ yields $\mathbf{A}'(\tau) = \boldsymbol{\iota} - \mathbf{M}\mathbf{A}(\tau)$ with $\mathbf{M}$ as in (43); the unique solution follows from initial conditions $\mathbf{A}(0) = \mathbf{0}$. □

Proof of Proposition 3. Substituting static relations (risk prices (40), effective rates (33), and the policy rules) into (25), (27), the dynamics of $(\check{\mathbf{c}}_t^\sigma, \mathbf{b}_t^\sigma)$ from Theorem 1, and (34) yields a linear expectational system in the jumps $\mathbf{y}_t$ and states $\mathbf{s}_t$ (Appendix B.3). Under standard saddle-path conditions, the unique bounded solution is $\mathbf{y}_t = \boldsymbol{\Omega}\mathbf{s}_t$; $\boldsymbol{\Gamma}$ is the implied equilibrium state dynamics; and $\boldsymbol{\sigma}_s$ is formed directly from $\boldsymbol{\sigma}_z$, $B\check{\boldsymbol{\Sigma}}$, and $B\boldsymbol{\eta}^\sigma + \boldsymbol{\sigma}_f$. □

Proof of Theorem 2. Given Prop. 3, every argument of (29)-(31) is linear in $\mathbf{s}_t$, so $\mathcal{L}_t = \mathbf{s}_t^\top \Psi \mathbf{s}_t$ with Ψ positive semi-definite. An equilibrium is a fixed point of $T(\,\cdot\,; a)$: loadings $\boldsymbol{\sigma}(\cdot) \mapsto$ composites (32) $\mapsto$ solution of the linear system (Prop. 3) $\mapsto$ implied loadings $-\mathbf{A}(\tau)^\top \boldsymbol{\sigma}_s$ (Prop. 2). At $a = 0$ risk prices drop out of the dynamics and the model is a determinate linear New Keynesian system: $T(\,\cdot\,; 0)$ is constant, so the fixed point exists and is unique; moreover, $\partial_\sigma T(\cdot, a)|_{a=0}$ exists and is well-behaved; the result follows from the implicit function theorem. □

Proof of Proposition 4. With $\beta(\tau) \equiv 0$ and no systematic balance sheet rule, (48) reduces to $\lambda_t = -\epsilon_{q,t}^\sigma - b_t^\sigma$. Conjecture $\lambda_t \equiv 0$: from (45), (47), the bounded solution from $b_0^\sigma = 0$ is $\check{c}_t^\sigma = b_t^\sigma = 0$, confirming the conjecture. All borrowing rates then equal i_t, hence the equilibrium corresponds to the representative-agent case. Then the Taylor rule and (46) give the results. A balance sheet shock enters directly: $\partial\lambda_t / \partial\epsilon_{q,t}^\sigma = -1$, the gap loading is $\eta^\sigma/(\phi_x + \varsigma\kappa_q) > 0$ at leading order in a, and $\lambda_t \not\equiv 0$ makes $\check{c}_t^\sigma$, hence b_{t+k}^σ, nonzero via (47). □

Proof of Proposition 5. (Impact.) At leading order in a, premium terms drop from (46): under $i_t = \phi_x x_t$, $x_t = \xi^0 z_t$ with $\xi^0 = v_z/(\phi_x + \varsigma\kappa_z)$, $v_z \equiv -\varsigma\kappa_z \frac{1+\varphi}{\varsigma+\varphi}$, and (48) gives $\lambda_t = \Lambda_z z_t$ with $\Lambda_z = \beta^\sigma \frac{1+\varphi}{\varsigma+\varphi} \cdot \frac{\phi_x - \bar{\phi}_x}{\phi_x + \varsigma\kappa_z} < 0$ iff $\phi_x < \bar{\phi}_x$: on impact $i_t = \phi_x x_t$ and λ_t decline together. *(Reversal.)* At next order in a, solving (47) gives $\check{c}_t^\sigma = \frac{\rho\varphi}{\varphi+\varsigma} b_t^\sigma + g z_t$, $g = -\check{\Sigma}\Lambda_z/(\varsigma(\rho + \kappa_z)) > 0$; in (45) the ρb^σ terms cancel, so $\mathrm{d}b_t^\sigma = -(1 + \varsigma/\varphi) g z_t \,\mathrm{d}t$ accumulates monotonically after a shock $z_0 < 0$. In $\lambda_t = \Lambda_z z_t - b_t^\sigma$ the first term decays at rate κ_z while the second grows, so λ_{t+k} crosses zero at a finite $\bar{k}$; meanwhile $x_{t+k} < 0$ at all horizons, so i_{t+k} stays below steady state. *(Balance sheet rule.)* $q e_t^\sigma = \phi_q^\sigma x_t$ with $\phi_q^\sigma < 0$ shifts the impact loading to $\Lambda_z - \phi_q^\sigma \xi^0 < \Lambda_z$, amplifying both the impact decline and the reversal. □

Proof of Corollary 5.1. Repeating the impact computation with $\epsilon_{i,t}$ (decay rate κ_i) in place of v_t: $x_t = \xi_\epsilon \epsilon_{i,t}$ with $\xi_\epsilon = -1/(\phi_x + \varsigma\kappa_i) < 0$, and λ_t loads as $\beta^\sigma(1 + \varsigma + \varphi)\xi_\epsilon < 0$ while the short rate loads as $1 + \phi_x \xi_\epsilon = \varsigma\kappa_i/(\phi_x + \varsigma\kappa_i) > 0$: a shock decreasing i_t raises x_t and λ_t. Given a unit change in the short rate, the conditional magnitude is $\beta^\sigma(1 + \varsigma + \varphi)/(\varsigma\kappa_i)$ against the unconditional ratio $\beta^\sigma(\bar{\phi}_x - \phi_x)/(\phi_x \varsigma\kappa_z)$ from Prop. 5; the former is smaller iff $\phi_x\,[1 + \kappa_z(1 + \varsigma + \varphi)/\kappa_i] < \bar{\phi}_x$. □

Proof of Lemma 1. At $B=0$, $\mathbf{X}_t \equiv \check{\mathbf{c}}_t^\sigma - \frac{\rho}{1+\varsigma/\varphi}\mathbf{b}_t^\sigma$ satisfies $\mathrm{E}_t\,\mathrm{d}\mathbf{X}_t = [\rho\mathbf{X}_t + \varsigma^{-1}a\check{\mathbf{\Sigma}}\mathbf{\Lambda}_t]\,\mathrm{d}t$; the unique bounded solution is $\mathbf{X}_t = -\varsigma^{-1}a\check{\mathbf{\Sigma}}\mathbf{P}_t$, which is (50). For (51), both $\mathbf{b}_t^\sigma$ and $\check{\mathbf{\Sigma}}\mathbf{P}_t$ lie in the range of $\check{\mathbf{\Sigma}}$; thus $\check{\mathbf{\Sigma}}\check{\mathbf{\Sigma}}^+\check{\mathbf{\Sigma}} = \check{\mathbf{\Sigma}}$; and in the affine equilibrium $\mathbf{P}_t = \mathbf{L}(\rho\mathbf{I}+\mathbf{\Gamma})^{-1}\mathbf{s}_t$. For (52): (i) gives $a\mathbf{P}_t = a\mathbf{\Lambda}_t/(\rho+\kappa_p)$, (ii) sets $\mathbf{b}^\sigma = 0$, and (iii) gives $a^2\mathbf{\Lambda}_t^\top\check{\mathbf{\Sigma}}\mathbf{\Lambda}_t = s_0^2\tilde{\lambda}_t^2$. □

Proof of Lemma 2. By Theorem 1, $\{qe_t(\tau)\}_{\tau=0}^T$ enters equilibrium conditions only through $\mathbf{qe}_t^\sigma$, and enters the loss (31) only through the holding cost, so the profile solves $\min_{\{qe_t(\tau)\}}\int_0^T \frac{\psi(\tau)}{2}qe(\tau)^2\,\mathrm{d}\tau$ subject to $\int_0^T \boldsymbol{\sigma}(\tau)^\top qe(\tau)\,\mathrm{d}\tau = \mathbf{qe}^\sigma$. Pointwise variation with multiplier $\mathbf{m}\in\mathbb{R}^{N_z}$ gives $\psi(\tau)qe(\tau) = \boldsymbol{\sigma}(\tau)\mathbf{m}$; the constraint then gives $\mathbf{m} = \mathbf{\Omega}_\psi^{-1}\mathbf{qe}^\sigma$, and the minimized cost follows. □

Proof of Proposition 6. *($\check{\Sigma}=0$.)* Zero exposure dispersion implies $\check{c}^\sigma = b^\sigma = 0$. Setting $x\equiv 0$ in (48) gives $\lambda_t = -\frac{\beta^\sigma}{\varsigma\kappa_z}v_t$, and (46) then requires $i_t = v_t - \eta^\sigma\lambda_t = \left[1+\frac{\eta^\sigma\beta^\sigma}{\varsigma\kappa_z}\right]v_t \equiv \chi_{i,v}^n v_t$, which implements $x\equiv 0$. *($\check{\Sigma}>0$.)* With $i_t = \chi_{i,v}v_t$, at leading order $x_t = \xi_v v_t$, $\xi_v = (1-\chi_{i,v})/(\varsigma\kappa_z)$, and $\lambda_v = \frac{\beta^\sigma}{\varsigma\kappa_z}[(1+\varsigma+\varphi)(1-\chi_{i,v})-1]$: 1 follows. For 2-3: as a function of $\chi_{i,v}^n$, $\mathrm{Var}[x_t]$ attains its minimum of zero, while the dispersion term $\propto \check{\Sigma}\lambda_v^2$ falls strictly as $\chi_{i,v}$ falls below $\chi_{i,v}^n$ (both directly and through the endogenous loadings, which scale with reactivity). Hence $\chi_{i,v}^* < \chi_{i,v}^n$, and at the optimum $x\not\equiv 0$, $\mathrm{Var}_i\,\check{c}_t(i)\neq 0$; b^σ non-negligible implies direct effects on λ_t directly; thus the optimal rule generically sets $\chi_{i,b}\neq 0$. □

Proof of Proposition 7. Conjecture $x\equiv 0$, $\lambda\equiv 0$. Given $x\equiv 0$, the balance sheet rule sets $\lambda_t = -\frac{\beta^\sigma}{\varsigma\kappa_z}v_t - qe_t^\sigma - b_t^\sigma = -b_t^\sigma$; then $\lambda\equiv 0$, implies $\check{c}_t^\sigma = \frac{\rho}{1+\varsigma/\varphi}b_t^\sigma$ and $\mathrm{d}b_t^\sigma = 0$, so initial equalization gives $b^\sigma = \check{c}^\sigma = 0$ for all t (confirming $\lambda\equiv 0$) and, household by household, $\check{c}_t(i) = 0$ (3). Then 2 follows from $\mu_t(\tau) - i_t = a\boldsymbol{\sigma}(\tau)\mathbf{\Lambda}_t = 0$; the IS curve with $\lambda = 0$, $i_t = v_t$ gives $x\equiv 0$ (i1); every term of (29)-(31) is zero. □

Proof of Proposition 8. *(Bond market stabilization.)* The rule sets $\lambda\equiv 0$; the argument of Prop. 7 gives zero dispersion, while (46) with $i_t = \tilde{\chi}_i v_t$ has bounded solution $x_t = \frac{1-\tilde{\chi}_i}{\varsigma\kappa_z}v_t \not\equiv 0$. *(Macro stabilization.)* $x\equiv 0$ requires $i_t + \eta^\sigma\lambda_t = v_t \iff \lambda_t = (1-\tilde{\chi}_i)v_t/\eta^\sigma \not\equiv 0$, implemented via (48) with $\chi_{q,v}^\sigma = -\frac{\beta^\sigma}{\varsigma\kappa_z} - \frac{1-\tilde{\chi}_i}{\eta^\sigma}$, $\chi_{q,b}^\sigma = -1$; then $\mathrm{E}_t\,\mathrm{d}\check{c}_t^\sigma = \varsigma^{-1}\check{\Sigma}\lambda_t\,\mathrm{d}t \not\equiv 0$. *(Impossibility and the optimum.)* $x\equiv 0$ and $\lambda\equiv 0$ jointly require $i_t = v_t$, contradicting $\tilde{\chi}_i\neq 1$; hence the optimum has $x\not\equiv 0$ and $\mathrm{Var}_i\,\check{c}_t(i)\neq 0$. For 1: interpolating rules with parameter θ, $\lambda_t = \theta(1-\tilde{\chi}_i)v_t/\eta^\sigma$ and $x_t = (1-\theta)\frac{1-\tilde{\chi}_i}{\varsigma\kappa_z}v_t$, so $\mathrm{Var}(x)\propto(1-\theta)^2$ while the dispersion loss $\propto\theta^2$: the marginal dispersion cost vanishes at $\theta=0$ while the marginal output gain does not, so $\theta^*>0$ and output gap volatility falls below the no-balance-sheet benchmark at leading order. □

Proof of Proposition 9. *(1.)* In any bounded equilibrium with $x\equiv\pi\equiv 0$, the IS curve gives $i_t + a\eta^\sigma\Lambda_t = v_t$ pointwise, and the Phillips curve gives $\kappa\alpha_{\hat{\mu}}\hat{\mu}_t = -u_t$ with $\kappa\alpha_{\hat{\mu}} = \epsilon/\vartheta$. That is, $\beta i_t + a\beta^\sigma\Lambda_t = -(\vartheta/\epsilon)u_t$, hence a 2×2 system in (i_t,Λ_t) with determinant

$a\Delta^\sigma \neq 0$, whose row reduction gives $a\Delta^\sigma \Lambda_t = -\nu_t$, i.e. (57); the qe rule reproduces the required Λ_t at $x = 0$, and conversely $(x, \pi) \equiv 0$ is the unique bounded solution under (57). *(2.)* The $-b_t^\sigma$ term in qe_t^σ cancels b^σ from Λ_t, so $\Lambda_t = L_z z_t$, $L_z = -\nu_z/(a\Delta^\sigma)$, and the dispersion block is triangular with eigenvalues $\{0, \rho\}$, hence boundedness implies $\check{c}_t^\sigma = \frac{\rho}{1+\varsigma/\varphi} b_t^\sigma + g_c z_t$, $g_c = \check{\Sigma}\nu_z/(\varsigma(\rho + \kappa_z)\Delta^\sigma)$, and $\mathrm{d}b_t^\sigma = -(1 + \varsigma/\varphi) g_c z_t \,\mathrm{d}t$. Thus with $b_0^\sigma = 0$, the unit root implies $\check{c}_t^\sigma = g_c(z_t - \rho \int_0^t z_s \,\mathrm{d}s)$, so the dispersion loss is $\propto \check{\Sigma}\nu_z^2/[(\rho + \kappa_z)^2(\Delta^\sigma)^2] \to \infty$ as $\Delta^\sigma \to 0$, provided $\nu_z \neq 0$ (that is, excluding the knife-edge case $u_z = -(\epsilon/\vartheta)\beta v_z$). The instrument loadings scale as $(\Delta^\sigma)^{-1}$, so they diverge as well. □

Proof of Lemma 3. Fixing bond volatility implies the composites are fixed, hence the problem is linear-quadratic with forward-looking constraints, and the optimality system is the continuous-time counterpart of the recursive-multiplier characterization of commitment problems (Marcet and Marimon 2019; Backus and Driffill 1986): the costates on the forward-looking equations serve as the planner's promise states, initialized at zero because the corresponding variables are free at $t = 0$, with the current-value drifts (61). In continuous time, these costates are locally deterministic: the jump variables' diffusions are free variables of the plan (steered by the innovation loadings of the announced instruments) and enter the Lagrangian only through their cross-variation with the multipliers, so optimality forces the multiplier diffusions to vanish. □

Proof of Theorem 3. Fix the volatility function $\{\boldsymbol{\sigma}(\tau)\}_{\tau=0}^\infty$, from which risk-weighted composites are computed; consistency of the plan with $\boldsymbol{\sigma}(\tau)$ implies the innovation loadings of (B5) must reproduce $\boldsymbol{\sigma}(\tau)$. This is the continuum of linear constraints (B6) on the plan's impulse-response kernels. Pairing these with multipliers $\boldsymbol{\nu}(\tau)$, the multipliers enter the objective only through $\mathrm{E}\left[i_t \upsilon_t^i + \boldsymbol{\Lambda}_t^\top \boldsymbol{\upsilon}_t^\Lambda\right]$ with the shadow factors (63); the augmented problem is the same as Lemma 3 plus a flow bilinear in the plan and $(\upsilon^i, \boldsymbol{\upsilon}^\Lambda)$. Thus, every derivative collecting the $\boldsymbol{\Lambda}$-risk-pricing-channels gains $\boldsymbol{\upsilon}_t^\Lambda$ additively, and the i-short-rate optimality conditions gain υ_t^i, with bounded solution linear in the extended state and in conditional expectations of the kernels in (66). Since the value depends on $\{\boldsymbol{\sigma}(\tau)\}_{\tau=0}^\infty$ only through c, stationarity in $\{\boldsymbol{\sigma}(\tau)\}_{\tau=0}^\infty$ is the linear Volterra equation (B7) with forcing $\sum_k \vartheta_k \partial c_k/\partial \boldsymbol{\sigma}(\tau)$; linearity delivers (64). Basis functions are computed in closed form (Appendix B.4). □

B Model Details

B.1 Household Optimality

Let $\zeta_t(i)$ denote the costate on nominal wealth, with $\mathrm{d}\zeta_t(i) = \zeta_t(i)[\mu_{\zeta,t}(i)\,\mathrm{d}t + \boldsymbol{\sigma}_{\zeta,t}(i)\,\mathrm{d}\mathbf{B}_t]$. The maximum principle gives the static conditions $e^{-\rho t + \varrho_t} C_t(i)^{-\varsigma} = \zeta_t(i)\mathcal{P}_t$ and $e^{-\rho t + \varrho_t} N_t(i)^\varphi = \zeta_t(i)\mathcal{W}_t$ and, since wealth enters the budget constraint linearly with

return $\mathrm{d}\tilde{R}_t(i)$, the costate condition $\mu_{\zeta,t}(i) = -[\tilde{\mu}_t(i) + \boldsymbol{\sigma}_{\zeta,t}(i)\tilde{\boldsymbol{\sigma}}_t(i)^\top]$. Applying Ito's lemma to $\zeta_t(i) = e^{-\rho t + \varrho_t} e^{-\varsigma c_t(i)} \mathcal{P}_t^{-1}$ (the price level carries no diffusion under Rotemberg pricing) and substituting yields the exact Euler equation

$$\mu_{c,t}(i) = \frac{1}{\varsigma}\left[\tilde{\mu}_t(i) - \pi_t - \rho - \kappa_\varrho \varrho_t\right] + \frac{\varsigma}{2}\left|\boldsymbol{\sigma}_{c,t}(i)\right|^2 - \boldsymbol{\sigma}_{c,t}(i)\tilde{\boldsymbol{\sigma}}_t(i)^\top + \frac{1}{\varsigma}\left[\tfrac{1}{2}\left|\boldsymbol{\sigma}_\varrho\right|^2 - \varsigma \boldsymbol{\sigma}_\varrho \boldsymbol{\sigma}_{c,t}(i)^\top + \boldsymbol{\sigma}_\varrho \tilde{\boldsymbol{\sigma}}_t(i)^\top\right], \quad \text{(B1)}$$

where the second line is equal to $\varsigma^{-1}\xi_t(i)$ in (9); and where $\mu_{c,t}(i)$ and $\boldsymbol{\sigma}_{c,t}(i)$ are the drift and diffusion of $c_t(i)$. Every term in the second lien is a product of two first-order diffusion loadings, hence second order under the perturbation of Section 2.2; to first order, $\mathrm{E}_t\,\mathrm{d}c_t(i) = \varsigma^{-1}\left[\tilde{\mu}_t(i) - \pi_t - \kappa_\varrho \varrho_t\right]\mathrm{d}t$ in deviations from steady state. The discount shock enters only through its expected change $-\kappa_\varrho \varrho_t$.

B.2 Firm Optimality and the Phillips Curve

Let $\gamma_t(j)$ be the costate on $\mathcal{P}_t(j)$ and $Q_t^{\mathcal{H}} \equiv \mathcal{P}_t^{-1}\int_0^1 C_t(i)^{-\varsigma}\,\mathrm{d}i$. Because the stochastic integral of the financing diffusion has zero expectation, only the drift of $\mathrm{d}\Pi_t(j)$ is priced, so the firm's flow objective depends on financing costs only through $\hat{\mu}_t$. The Hamiltonian is $e^{-\rho t}Q_t^{\mathcal{H}}F + \gamma_t(j)\mathcal{P}_t(j)\pi_t(j)$, with first-order condition $\gamma_t(j)\mathcal{P}_t(j) = e^{-\rho t}Q_t^{\mathcal{H}}\vartheta\pi_t(j)\mathcal{P}_t Y_t$ and costate equation $\mathrm{E}_t\,\mathrm{d}\gamma_t(j) = -[e^{-\rho t}Q_t^{\mathcal{H}}F_{\mathcal{P}(j)} + \gamma_t(j)\pi_t(j)]\,\mathrm{d}t$. Differentiating and substituting along with symmetry gives (15).

Linearization. Around the zero-inflation steady state, $\mathrm{E}_t\,\mathrm{d}\left(Q^{\mathcal{H}}Y\pi\right)/(Q^{\mathcal{H}}Y) = \mathrm{E}_t\,\mathrm{d}\pi + O(\xi^2)$, and log real marginal cost is $mc_t = (\varsigma+\varphi)x_t + \frac{\hat{\mu}_t + \rho\hat{\beta}_t}{1+\beta\rho}$, using $w_t - z_t = (\varsigma+\varphi)x_t$ and splitting the financing rate into the portfolio component $\hat{\mu}_t$ (steady-state weights) and the financing shock component $\rho\hat{\beta}_t$. Substituting into (15) gives (27) with

$$\kappa = \frac{\epsilon(1+\beta\rho)(\varsigma+\varphi)}{\vartheta}, \quad u_t = -\frac{1+\tau^y}{\vartheta\,\epsilon}\left(\epsilon_t - \epsilon\right) = -\frac{1+\beta\rho}{\vartheta\,(\epsilon-1)}\left(\epsilon_t - \epsilon\right). \quad \text{(B2)}$$

Note that a fall in the demand elasticity is interpreted as a rise in the desired markup. Since $\mathrm{E}\,\epsilon_t = \epsilon$ and ϵ_t enters the welfare derivation only through the symmetric-equilibrium aggregator, markup shocks leave the steady state, the subsidy (A1), and the form of the loss (29)-(31) unchanged.

B.3 The First-Order Equilibrium System

Eliminating the static relations (risk prices (Theorem 1.4), effective rates (33), financing shocks, and the policy rules $i_t = \boldsymbol{\phi}_i^\top \mathbf{s}_t$, $\mathbf{qe}_t^\sigma = \boldsymbol{\Phi}_q \mathbf{s}_t$) gives dynamics

$$\begin{aligned}
\mathrm{E}_t\, \mathrm{d}x_t &= \varsigma^{-1} \left[i_t + a\boldsymbol{\eta}^\sigma \boldsymbol{\Lambda}_t - \pi_t - v_t \right] \mathrm{d}t\,, \\
\mathrm{E}_t\, \mathrm{d}\pi_t &= \left[\rho\pi_t - \kappa \left(x_t + \alpha_{\hat{\mu}} \hat{\mu}_t + \alpha_{\hat{\beta}} \hat{\beta}_t \right) - u_t \right] \mathrm{d}t\,, \\
\mathrm{E}_t\, \mathrm{d}\check{\mathbf{c}}_t^\sigma &= \varsigma^{-1} a\, \check{\boldsymbol{\Sigma}} \boldsymbol{\Lambda}_t \, \mathrm{d}t\,, \\
\mathrm{d}\mathbf{b}_t^\sigma &= \left[\rho \mathbf{b}_t^\sigma + aB\check{\boldsymbol{\Sigma}} \boldsymbol{\Lambda}_t - (1 + \varsigma/\varphi)\, \check{\mathbf{c}}_t^\sigma \right] \mathrm{d}t + B\check{\boldsymbol{\Sigma}}\, \mathrm{d}\mathbf{B}_t\,, \\
\mathrm{d}b_t &= \left[\rho b_t + B\left(\tilde{\mu}_t - \pi_t \right) + (\varsigma + \varphi) x_t + f_t \right] \mathrm{d}t + \left[B\boldsymbol{\eta}^\sigma + \boldsymbol{\sigma}_f \right] \mathrm{d}\mathbf{B}_t\,,
\end{aligned} \tag{B3}$$

together with the factor dynamics (1), where $\boldsymbol{\Lambda}_t$ and $\hat{\mu}_t$ are the linear functions of $(\mathbf{y}_t, \mathbf{s}_t)$ given by (40) and (33). Stacking the jump variables $\mathbf{y}_t = (x_t, \pi_t, \check{\mathbf{c}}_t^{\sigma\top})^\top$ ($2 + N_z$ variables) on the states $\mathbf{s}_t = (\mathbf{z}_t^\top, \mathbf{b}_t^{\sigma\top}, b_t)^\top$ ($2N_z + 1$ variables, plus any policy states) yields

$$\begin{pmatrix} \mathrm{E}_t\, \mathrm{d}\mathbf{y}_t \,/\, \mathrm{d}t \\ \mathrm{E}_t\, \mathrm{d}\mathbf{s}_t \,/\, \mathrm{d}t \end{pmatrix} = \mathbf{H} \begin{pmatrix} \mathbf{y}_t \\ \mathbf{s}_t \end{pmatrix}, \qquad \boldsymbol{\sigma}_s = \begin{pmatrix} \boldsymbol{\sigma}_z \\ B\check{\boldsymbol{\Sigma}} \\ B\boldsymbol{\eta}^\sigma + \boldsymbol{\sigma}_f \end{pmatrix}, \tag{B4}$$

where $\mathbf{H}$ collects the coefficients of (B3) after the static substitutions (a known function of deep parameters, the composites (32), the shock loading maps, and the policy coefficients). Assume standard saddle-path conditions: exactly $2 + N_z$ eigenvalues of $\mathbf{H}$ with positive real part; when $B \neq 0$ the wealth states are counted with the predetermined variables, as their innovations are prescribed revaluations. Then the unique bounded solution is $\mathbf{y}_t = \boldsymbol{\Omega}\mathbf{s}_t$, and $\boldsymbol{\Gamma}$ in (44) is the implied equilibrium dynamics matrix.

Fixed-point algorithm. (1) Guess loadings $\{\boldsymbol{\sigma}(\tau)\}_{\tau=0}^{\infty}$; (2) form the composites (32) and risk-weighted policy terms; (3) solve (B4) for $(\boldsymbol{\Omega}, \boldsymbol{\Gamma}, \boldsymbol{\iota}, \mathbf{L})$; (4) update $\mathbf{A}(\tau)$ from Prop. 2 and set $\boldsymbol{\sigma}(\tau) = -\mathbf{A}(\tau)^\top \boldsymbol{\sigma}_s$; (5) iterate to convergence. Theorem 2 guarantees a locally unique fixed point near $a = 0$; away from that neighborhood, we select the solution continuously connected in a to the $a = 0$ equilibrium; this same homotopy simultaneously serves as the solution algorithm and the selection device.

B.4 Full Commitment: Computation

Fix a candidate loading function $\boldsymbol{\sigma}(\cdot)$; every composite is computed from it directly. Given $\boldsymbol{\sigma}(\cdot)$, integrating the arbitrageur condition at first order gives

$$\log P_t^{(\tau)} = -\rho\tau - \mathrm{E}_t \int_0^\tau \left[i_{t+u} + a\boldsymbol{\sigma}(\tau - u) \boldsymbol{\Lambda}_{t+u} \right] \mathrm{d}u\,, \tag{B5}$$

so that, writing $\mathbf{k}_y(u)$ for the impulse-response kernel of a process y (the response of $\mathrm{E}_t[y_{t+u}]$ to $\mathrm{d}\mathbf{B}_t$), consistency of the plan with the candidate loadings requires

$$\boldsymbol{\sigma}(\tau) = -\int_0^\tau \left[\mathbf{k}_i(u) + a\,\boldsymbol{\sigma}(\tau-u)\,\mathbf{k}_\Lambda(u)\right]\mathrm{d}u \qquad \forall \tau, \tag{B6}$$

a continuum of linear equality constraints on the plan (equivalent to the fixed point of Prop. 2). Pairing (B6) with multipliers $\boldsymbol{\nu}(\tau)$ and interchanging the order of integration produces the shadow factors (63). Differentiating the Lagrangian in $\boldsymbol{\sigma}(\tau)$ gives the linear Volterra equation of the second kind

$$\boldsymbol{\nu}^{(k)}(\tau) + a\int_\tau^T \mathbf{k}_\Lambda(s-\tau)^\top \boldsymbol{\nu}^{(k)}(s)\,\mathrm{d}s = \mathbf{w}_k(\tau) \equiv \partial c_k/\partial \boldsymbol{\sigma}(\tau)^\top, \tag{B7}$$

uniquely solvable in a single backward pass from the long end ($\boldsymbol{\nu}^{(k)}(T) = \mathbf{w}_k(T)$). The substitution $\boldsymbol{\phi}(\tau) \equiv \int_\tau^T e^{-\boldsymbol{\Gamma}^{+\top}(s-\tau)}\mathbf{L}^{+\top}\boldsymbol{\nu}^{(k)}(s)\,\mathrm{d}s$ converts (B7) into the adjoint of the pricing ODE, with closed form

$$\boldsymbol{\nu}^{(k)}(\tau) = \mathbf{w}_k(\tau) - a\,\boldsymbol{\Sigma}_s^{+\top}\int_\tau^T e^{\mathbf{M}^{+\top}(\tau-s)}\,\mathbf{L}^{+\top}\mathbf{w}_k(s)\;\mathrm{d}s\,, \tag{B8}$$

where $+$ superscripts denote extended-state objects. *Algorithm* (fixed point in $(c,\boldsymbol{\vartheta}) \in \mathbb{R}^{2K}$): given $(c,\boldsymbol{\vartheta})$, (1) solve the K basis equations (B8) and form $\boldsymbol{\nu} = \sum_k \vartheta_k \boldsymbol{\nu}^{(k)}$ and the shadow kernels (63); (2) solve the amended linear optimality system of Theorem 3 for the plan; (3) update $\boldsymbol{\sigma}(\cdot)$, hence c; (4) update $\boldsymbol{\vartheta} = \partial J/\partial c$ from envelope formulas (stationary bilinear moments of the solved system). Every step is a linear solve or an explicit formula; setting $\boldsymbol{\vartheta} = 0$ reproduces the plan of Lemma 3. When balance sheet costs are specified directly on the risk-weighted instrument (an exogenous quadratic cost), the induced cost metric is invariant to the loadings and drops from the composite list; otherwise c also includes the entries of $\boldsymbol{\Omega}_\psi \equiv \int_0^T \boldsymbol{\sigma}(\tau)^\top\boldsymbol{\sigma}(\tau)/\psi(\tau)\,\mathrm{d}\tau$.

B.5 Microfoundation of Fixed Portfolio Weights

Fixed heterogeneous weights are individually optimal under even small fixed rebalancing costs, because portfolio choice is a *second-order* problem (and this remains second-order even with the perturbation approach utilized in this paper, since households are not directly affected by the arbitrageur frictions). With $B = 0$, relative wealth is locally deterministic, so $\mathrm{d}(e^{-\rho t}\check{b}_t(i)) = -e^{-\rho t}(1+\varsigma/\varphi)\check{c}_t(i)\,\mathrm{d}t$ holds under any portfolio weight choice, even with time-variation in weights. Additionally, $\int_0^\infty e^{-\rho t}(1+\varsigma/\varphi)\check{c}_t(i)\,\mathrm{d}t = \check{b}_0(i)$, so utility is independent to portfolio weights at first order, and utility differences across weight policies are $O(\xi^2)$. Hence for any fixed cost $\varkappa > 0$ there exists $\bar{\xi}(\varkappa) > 0$ such that for all $\xi < \bar{\xi}$, never rebalancing is optimal for every household. However, while each individual household's loss due to fixed

rebalancing costs is second order, the dispersion term (30) is the population sum of losses across all individuals; policy that stabilizes term premia removes these costs without individual households paying the fixed cost.